

AN IMAGE IS WORTH 16X16 WORDS: TRANSFORMERS FOR IMAGE RECOGNITION AT SCALE

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ABSTRACT

While the Transformer architecture has become the de-facto standard for natural language processing tasks, its applications to computer vision remain limited. In vision, attention is either applied in conjunction with convolutional networks, or used to replace certain components of convolutional networks while keeping their overall structure in place. We show that this reliance on CNNs is not necessary and a pure transformer applied directly to sequences of image patches can perform very well on image classification tasks. When pre-trained on large amounts of data and transferred to multiple mid-sized or small image recognition benchmarks (ImageNet, CIFAR-100, VTAB, etc.), Vision Transformer (ViT) attains excellent results compared to state-of-the-art convolutional networks while requiring substantially fewer computational resources to train.¹

1 INTRODUCTION

Self-attention-based architectures, in particular Transformers (Vaswani et al., 2017), have become the model of choice in natural language processing (NLP). The dominant approach is to pre-train on a large text corpus and then fine-tune on a smaller task-specific dataset (Devlin et al., 2019). Thanks to Transformers’ computational efficiency and scalability, it has become possible to train models of unprecedented size, with over 100B parameters (Brown et al., 2020; Lepikhin et al., 2020). With the models and datasets growing, there is still no sign of saturating performance.

In computer vision, however, convolutional architectures remain dominant (LeCun et al., 1989; Krizhevsky et al., 2012; He et al., 2016). Inspired by NLP successes, multiple works try combining CNN-like architectures with self-attention (Wang et al., 2018; Carion et al., 2020), some replacing the convolutions entirely (Ramachandran et al., 2019; Wang et al., 2020a). The latter models, while theoretically efficient, have not yet been scaled effectively on modern hardware accelerators due to the use of specialized attention patterns. Therefore, in large-scale image recognition, classic ResNet-like architectures are still state of the art (Mahajan et al., 2018; Xie et al., 2020; Kolesnikov et al., 2020).

Inspired by the Transformer scaling successes in NLP, we experiment with applying a standard Transformer directly to images, with the fewest possible modifications. To do so, we split an image into patches and provide the sequence of linear embeddings of these patches as an input to a Transformer. Image patches are treated the same way as tokens (words) in an NLP application. We train the model on image classification in supervised fashion.

When trained on mid-sized datasets such as ImageNet without strong regularization, these models yield modest accuracies of a few percentage points below ResNets of comparable size. This seemingly discouraging outcome may be expected: Transformers lack some of the inductive biases

¹Fine-tuning code and pre-trained models are available at https://github.com/google-research/vision_transformer

一张图像值16x16个词：大规模图像识别的Transformer模型

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摘要

尽管Transformer架构已成为自然语言处理任务的事实标准，但其在计算机视觉中的应用仍然有限。在视觉领域，注意力机制要么与卷积网络结合使用，要么用于替换卷积网络的某些组件，同时保持其整体结构不变。我们证明这种对卷积网络的依赖并非必要，直接应用于图像块序列的纯Transformer模型能在图像分类任务上表现出色。当在大规模数据上进行预训练并迁移至多个中型或小型图像识别基准测试（如ImageNet、CIFAR-100、VTAB等）时，视觉Transformer（ViT）相比最先进的卷积网络取得了优异成果，同时训练所需的计算资源显著减少。¹

1 引言

基于自注意力的架构，特别是Transformer（Vaswani等人，2017），已成为自然语言处理（NLP）中的首选模型。主流方法是在大型文本语料库上进行预训练，然后在较小的任务特定数据集上进行微调（Devlin等人，2019）。得益于Transformer的计算效率和可扩展性，训练具有超过1000亿参数的史无前例的大规模模型已成为可能（Brown等人，2020；Lepikhin等人，2020）。随着模型和数据集的不断增长，性能提升仍未显现饱和迹象。

然而在计算机视觉领域，卷积架构仍占据主导地位（LeCun et al., 1989; Krizhevsky et al., 2012; He et al., 2016）。受自然语言处理领域成果的启发，多项研究尝试将类CNN架构与自注意力机制结合（Wang et al., 2018; Carion et al., 2020），部分研究甚至完全取代了卷积操作（Ramachandran et al., 2019; Wang et al., 2020a）。后一类模型虽在理论上高效，但由于采用了特殊的注意力模式，尚未在现代硬件加速器上实现有效的规模化应用。因此在大规模图像识别任务中，经典的类ResNet架构仍保持着最先进的性能（Mahajan et al., 2018; Xie et al., 2020; Kolesnikov et al., 2020）。

受Transformer在NLP领域规模化成功的启发，我们尝试将标准Transformer以最少的修改直接应用于图像。为此，我们将图像分割为多个图块，并将这些图块的线性嵌入序列作为输入提供给Transformer。图像块的处理方式与NLP应用中的标记（单词）相同。我们以监督学习方式训练该模型进行图像分类任务。

当在没有强正则化的中等规模数据集（如ImageNet）上训练时，这些模型的准确率仅比同等规模的ResNets低几个百分点，表现平平。这种看似令人失望的结果或许在意料之中：Transformer模型缺乏某些归纳偏置。

¹Fine-tuning code and pre-trained models are available at https://github.com/google-research/vision_transformer

inherent to CNNs, such as translation equivariance and locality, and therefore do not generalize well when trained on insufficient amounts of data.

However, the picture changes if the models are trained on larger datasets (14M-300M images). We find that large scale training trumps inductive bias. Our Vision Transformer (ViT) attains excellent results when pre-trained at sufficient scale and transferred to tasks with fewer datapoints. When pre-trained on the public ImageNet-21k dataset or the in-house JFT-300M dataset, ViT approaches or beats state of the art on multiple image recognition benchmarks. In particular, the best model reaches the accuracy of 88.55% on ImageNet, 90.72% on ImageNet-ReaL, 94.55% on CIFAR-100, and 77.63% on the VTAB suite of 19 tasks.

2 RELATED WORK

Transformers were proposed by Vaswani et al. (2017) for machine translation, and have since become the state of the art method in many NLP tasks. Large Transformer-based models are often pre-trained on large corpora and then fine-tuned for the task at hand: BERT (Devlin et al., 2019) uses a denoising self-supervised pre-training task, while the GPT line of work uses language modeling as its pre-training task (Radford et al., 2018; 2019; Brown et al., 2020).

Naive application of self-attention to images would require that each pixel attends to every other pixel. With quadratic cost in the number of pixels, this does not scale to realistic input sizes. Thus, to apply Transformers in the context of image processing, several approximations have been tried in the past. Parmar et al. (2018) applied the self-attention only in local neighborhoods for each query pixel instead of globally. Such local multi-head dot-product self attention blocks can completely replace convolutions (Hu et al., 2019; Ramachandran et al., 2019; Zhao et al., 2020). In a different line of work, Sparse Transformers (Child et al., 2019) employ scalable approximations to global self-attention in order to be applicable to images. An alternative way to scale attention is to apply it in blocks of varying sizes (Weissenborn et al., 2019), in the extreme case only along individual axes (Ho et al., 2019; Wang et al., 2020a). Many of these specialized attention architectures demonstrate promising results on computer vision tasks, but require complex engineering to be implemented efficiently on hardware accelerators.

Most related to ours is the model of Cordonnier et al. (2020), which extracts patches of size 2×2 from the input image and applies full self-attention on top. This model is very similar to ViT, but our work goes further to demonstrate that large scale pre-training makes vanilla transformers competitive with (or even better than) state-of-the-art CNNs. Moreover, Cordonnier et al. (2020) use a small patch size of 2×2 pixels, which makes the model applicable only to small-resolution images, while we handle medium-resolution images as well.

There has also been a lot of interest in combining convolutional neural networks (CNNs) with forms of self-attention, e.g. by augmenting feature maps for image classification (Bello et al., 2019) or by further processing the output of a CNN using self-attention, e.g. for object detection (Hu et al., 2018; Carion et al., 2020), video processing (Wang et al., 2018; Sun et al., 2019), image classification (Wu et al., 2020), unsupervised object discovery (Locatello et al., 2020), or unified text-vision tasks (Chen et al., 2020c; Lu et al., 2019; Li et al., 2019).

Another recent related model is image GPT (iGPT) (Chen et al., 2020a), which applies Transformers to image pixels after reducing image resolution and color space. The model is trained in an unsupervised fashion as a generative model, and the resulting representation can then be fine-tuned or probed linearly for classification performance, achieving a maximal accuracy of 72% on ImageNet.

Our work adds to the increasing collection of papers that explore image recognition at larger scales than the standard ImageNet dataset. The use of additional data sources allows to achieve state-of-the-art results on standard benchmarks (Mahajan et al., 2018; Touvron et al., 2019; Xie et al., 2020). Moreover, Sun et al. (2017) study how CNN performance scales with dataset size, and Kolesnikov et al. (2020); Djolonga et al. (2020) perform an empirical exploration of CNN transfer learning from large scale datasets such as ImageNet-21k and JFT-300M. We focus on these two latter datasets as well, but train Transformers instead of ResNet-based models used in prior works.

CNN固有的特性，如平移等变性和局部性，因此在数据量不足的情况下训练时泛化能力不佳。

然而，如果模型在更大的数据集（1400万至3亿张图像）上进行训练，情况就会发生变化。我们发现，大规模训练能够超越归纳偏置。当我们的视觉变换器（ViT）在足够规模上进行预训练并迁移到数据点较少的任务时，取得了优异的结果。在公共的ImageNet-21k数据集或内部的JFT-300M数据集上进行预训练后，ViT在多个图像识别基准测试中接近或超越了现有技术水平。具体而言，最佳模型在ImageNet上达到了88.55%的准确率，在ImageNet-Real上达到90.72%，在CIFAR-100上达到94.55%，在包含19个任务的VTAB套件上达到77.63%。

2 相关工作

Transformer由Vaswani等人（2017年）针对机器翻译任务提出，随后成为众多自然语言处理任务中的前沿方法。基于Transformer的大型模型通常在大规模语料上进行预训练，随后针对具体任务进行微调：BERT（Devlin等人，2019年）采用去噪自监督预训练任务，而GPT系列工作则以语言建模作为预训练任务（Radford等人，2018年；2019年；Brown等人，2020年）。

将自注意力机制直接应用于图像处理时，每个像素都需要关注所有其他像素。由于计算成本与像素数量呈二次方增长，这种方法难以扩展到实际输入尺寸。因此，为在图像处理中应用Transformer模型，过去已尝试多种近似方案。Parmar等人（2018）将自注意力限制在每个查询像素的局部邻域内而非全局范围。这类局部多头点积自注意力模块可完全替代卷积操作（Hu等人，2019；Ramachandran等人，2019；Zhao等人，2020）。另一研究方向中，稀疏Transformer（Child等人，2019）采用可扩展的全局自注意力近似方法以适应图像处理需求。另一种扩展注意力机制的方式是将其应用于不同尺寸的区块（Weissenborn等人，2019），极端情况下仅沿单个轴向进行注意力计算（Ho等人，2019；Wang等人，2020a）。尽管这些专用注意力架构在计算机视觉任务中展现出潜力，但其在硬件加速器上的高效实现仍需复杂的工程优化。

与我们的工作最相关的是Cordonnier等人（2020）提出的模型，该模型从输入图像中提取尺寸为 2×2 的图像块，并在其基础上应用完整的自注意力机制。该模型与ViT非常相似，但我们的研究进一步证明，大规模预训练能使原始Transformer模型与最先进的CNN相竞争（甚至更优）。此外，Cordonnier等人（2020）使用了 2×2 像素的小尺寸图像块，这导致模型仅适用于低分辨率图像，而我们的方法还能处理中等分辨率图像。

将卷积神经网络（CNNs）与自注意力机制相结合也引起了广泛关注，例如通过增强图像分类的特征图（Bello等人，2019），或利用自注意力进一步处理CNN的输出，应用于目标检测（Hu等人，2018；Carion等人，2020）、视频处理（Wang等人，2018；Sun等人，2019）、图像分类（Wu等人，2020）、无监督目标发现（Locatello等人，2020）或统一的文本-视觉任务（Chen等人，2020c；Lu等人，2019；Li等人，2019）。

另一个最近的相关模型是图像GPT（iGPT）（Chen等人，2020a），该模型在降低图像分辨率和色彩空间后，将Transformer应用于图像像素。该模型以无监督方式作为生成模型进行训练，所得表征随后可通过微调或线性探测来评估分类性能，在ImageNet上达到了72%的最高准确率。

我们的工作为日益增多的探索超越标准ImageNet数据集规模图像识别的研究文献增添了新内容。利用额外数据源能够在标准基准测试中取得最先进的结果（Mahajan等人，2018；Touvron等人，2019；Xie等人，2020）。此外，Sun等人（2017）研究了CNN性能如何随数据集规模扩展，而Kolesnikov等人（2020）与Djolonga等人（2020）则对基于ImageNet-21k和JFT-300M等大规模数据集的CNN迁移学习进行了实证探索。我们同样聚焦于后两个数据集，但采用Transformer模型进行训练，而非先前研究中基于ResNet的架构。

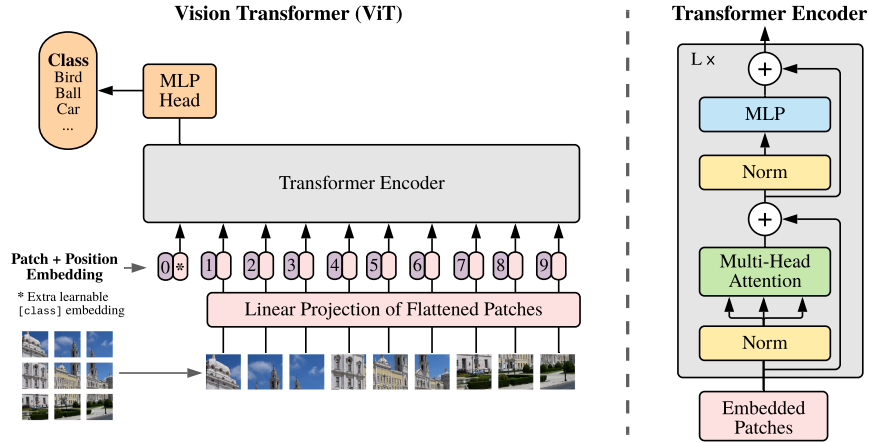


Figure 1: Model overview. We split an image into fixed-size patches, linearly embed each of them, add position embeddings, and feed the resulting sequence of vectors to a standard Transformer encoder. In order to perform classification, we use the standard approach of adding an extra learnable “classification token” to the sequence. The illustration of the Transformer encoder was inspired by Vaswani et al. (2017).

3 METHOD

In model design we follow the original Transformer (Vaswani et al., 2017) as closely as possible. An advantage of this intentionally simple setup is that scalable NLP Transformer architectures – and their efficient implementations – can be used almost out of the box.

3.1 VISION TRANSFORMER (ViT)

An overview of the model is depicted in Figure 1. The standard Transformer receives as input a 1D sequence of token embeddings. To handle 2D images, we reshape the image $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$ into a sequence of flattened 2D patches $\mathbf{x}_p \in \mathbb{R}^{N \times (P^2 \cdot C)}$, where (H, W) is the resolution of the original image, C is the number of channels, (P, P) is the resolution of each image patch, and $N = HW/P^2$ is the resulting number of patches, which also serves as the effective input sequence length for the Transformer. The Transformer uses constant latent vector size D through all of its layers, so we flatten the patches and map to D dimensions with a trainable linear projection (Eq. 1). We refer to the output of this projection as the patch embeddings.

Similar to BERT’s [class] token, we prepend a learnable embedding to the sequence of embedded patches ($\mathbf{z}_0^0 = \mathbf{x}_{\text{class}}$), whose state at the output of the Transformer encoder ($\mathbf{z}_L^0$) serves as the image representation $\mathbf{y}$ (Eq. 4). Both during pre-training and fine-tuning, a classification head is attached to $\mathbf{z}_L^0$. The classification head is implemented by a MLP with one hidden layer at pre-training time and by a single linear layer at fine-tuning time.

Position embeddings are added to the patch embeddings to retain positional information. We use standard learnable 1D position embeddings, since we have not observed significant performance gains from using more advanced 2D-aware position embeddings (Appendix D.4). The resulting sequence of embedding vectors serves as input to the encoder.

The Transformer encoder (Vaswani et al., 2017) consists of alternating layers of multiheaded self-attention (MSA, see Appendix A) and MLP blocks (Eq. 2, 3). Layernorm (LN) is applied before every block, and residual connections after every block (Wang et al., 2019; Baevski & Auli, 2019).

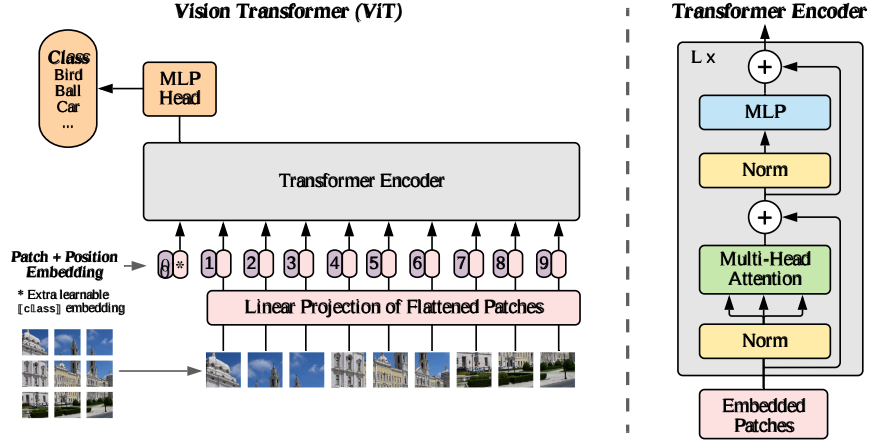


图1: 模型概览。我们将图像分割为固定大小的图像块, 对每个图像块进行线性嵌入, 添加位置嵌入, 并将得到的向量序列输入标准Transformer编码器。为执行分类任务, 我们采用标准方法, 在序列中添加一个额外的可学习“分类标记”。本Transformer编码器示意图的绘制灵感来源于Vaswani等人(2017)的研究。

3 方法

在模型设计中, 我们尽可能严格遵循原始Transformer (Vaswani等人, 2017) 的架构。这种有意简化的设置有一个优势: 可扩展的NLP Transformer架构及其高效实现几乎可以直接使用。

3.1 视觉变换器 (ViT)

模型的概览如图1所示。标准Transformer接收一维的令牌嵌入序列作为输入。为了处理二维图像, 我们将图像 $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$ 重塑为一系列展平的二维图像块 $\mathbf{x}_p \in \mathbb{R}^{N \times (P^2 \cdot C)}$, 其中 (H, W) 是原始图像的分辨率, C 是通道数, (P, P) 是每个图像块的分辨率, $N = HW/P^2$ 是生成的图像块数量, 这也作为Transformer的有效输入序列长度。Transformer在所有层中使用恒定的潜在向量大小 D , 因此我们将图像块展平并通过可训练的线性投影映射到 D 维 (公式1)。我们将此投影的输出称为图像块嵌入。

类似于BERT的[class]标记, 我们在嵌入补丁序列 ($\mathbf{z}_0^0 = \mathbf{x}_{\text{class}}$) 前添加了一个可学习的嵌入, 其在Transformer编码器输出端 ($\mathbf{z}_L^0$) 的状态作为图像表示 $\mathbf{y}$ (式4)。在预训练和微调阶段, 分类头均被附加到 $\mathbf{z}_L^0$ 。该分类头在预训练时由包含一个隐藏层的MLP实现, 在微调时则由单一线性层实现。

位置嵌入被添加到补丁嵌入中以保留位置信息。我们使用标准的可学习一维位置嵌入, 因为我们没有观察到使用更先进的二维感知位置嵌入会带来显著的性能提升 (附录D.4)。生成的嵌入向量序列将作为编码器的输入。

Transformer编码器 (Vaswani等人, 2017) 由多头自注意力模块 (MSA, 见附录A) 与MLP块交替堆叠而成 (公式2、3)。每个模块前应用层归一化 (LN), 每个模块后添加残差连接 (Wang等人, 2019; Baevski & Auli, 2019)。

The MLP contains two layers with a GELU non-linearity.

$$\mathbf{z}_0 = [\mathbf{x}_{\text{class}}; \mathbf{x}_p^1 \mathbf{E}; \mathbf{x}_p^2 \mathbf{E}; \dots; \mathbf{x}_p^N \mathbf{E}] + \mathbf{E}_{\text{pos}}, \quad \mathbf{E} \in \mathbb{R}^{(P^2 \cdot C) \times D}, \mathbf{E}_{\text{pos}} \in \mathbb{R}^{(N+1) \times D} \quad (1)$$

$$\mathbf{z}'_\ell = \text{MSA}(\text{LN}(\mathbf{z}_{\ell-1})) + \mathbf{z}_{\ell-1}, \quad \ell = 1 \dots L \quad (2)$$

$$\mathbf{z}_\ell = \text{MLP}(\text{LN}(\mathbf{z}'_\ell)) + \mathbf{z}'_\ell, \quad \ell = 1 \dots L \quad (3)$$

$$\mathbf{y} = \text{LN}(\mathbf{z}_L^0) \quad (4)$$

Inductive bias. We note that Vision Transformer has much less image-specific inductive bias than CNNs. In CNNs, locality, two-dimensional neighborhood structure, and translation equivariance are baked into each layer throughout the whole model. In ViT, only MLP layers are local and translationally equivariant, while the self-attention layers are global. The two-dimensional neighborhood structure is used very sparingly: in the beginning of the model by cutting the image into patches and at fine-tuning time for adjusting the position embeddings for images of different resolution (as described below). Other than that, the position embeddings at initialization time carry no information about the 2D positions of the patches and all spatial relations between the patches have to be learned from scratch.

Hybrid Architecture. As an alternative to raw image patches, the input sequence can be formed from feature maps of a CNN (LeCun et al., 1989). In this hybrid model, the patch embedding projection $\mathbf{E}$ (Eq. 1) is applied to patches extracted from a CNN feature map. As a special case, the patches can have spatial size 1×1 , which means that the input sequence is obtained by simply flattening the spatial dimensions of the feature map and projecting to the Transformer dimension. The classification input embedding and position embeddings are added as described above.

3.2 FINE-TUNING AND HIGHER RESOLUTION

Typically, we pre-train ViT on large datasets, and fine-tune to (smaller) downstream tasks. For this, we remove the pre-trained prediction head and attach a zero-initialized $D \times K$ feedforward layer, where K is the number of downstream classes. It is often beneficial to fine-tune at higher resolution than pre-training (Touvron et al., 2019; Kolesnikov et al., 2020). When feeding images of higher resolution, we keep the patch size the same, which results in a larger effective sequence length. The Vision Transformer can handle arbitrary sequence lengths (up to memory constraints), however, the pre-trained position embeddings may no longer be meaningful. We therefore perform 2D interpolation of the pre-trained position embeddings, according to their location in the original image. Note that this resolution adjustment and patch extraction are the only points at which an inductive bias about the 2D structure of the images is manually injected into the Vision Transformer.

4 EXPERIMENTS

We evaluate the representation learning capabilities of ResNet, Vision Transformer (ViT), and the hybrid. To understand the data requirements of each model, we pre-train on datasets of varying size and evaluate many benchmark tasks. When considering the computational cost of pre-training the model, ViT performs very favourably, attaining state of the art on most recognition benchmarks at a lower pre-training cost. Lastly, we perform a small experiment using self-supervision, and show that self-supervised ViT holds promise for the future.

4.1 SETUP

Datasets. To explore model scalability, we use the ILSVRC-2012 ImageNet dataset with 1k classes and 1.3M images (we refer to it as ImageNet in what follows), its superset ImageNet-21k with 21k classes and 14M images (Deng et al., 2009), and JFT (Sun et al., 2017) with 18k classes and 303M high-resolution images. We de-duplicate the pre-training datasets w.r.t. the test sets of the downstream tasks following Kolesnikov et al. (2020). We transfer the models trained on these dataset to several benchmark tasks: ImageNet on the original validation labels and the cleaned-up Real labels (Beyer et al., 2020), CIFAR-10/100 (Krizhevsky, 2009), Oxford-IIIT Pets (Parkhi et al., 2012), and Oxford Flowers-102 (Nilsback & Zisserman, 2008). For these datasets, pre-processing follows Kolesnikov et al. (2020).

MLP包含两层，并采用GELU非线性激活函数。

$$\mathbf{z}_0 = [\mathbf{x}_{\text{class}}; \mathbf{x}_p^1 \mathbf{E}; \mathbf{x}_p^2 \mathbf{E}; \dots; \mathbf{x}_p^N \mathbf{E}] + \mathbf{E}_{\text{pos}}, \quad \mathbf{E} \in \mathbb{R}^{(P^2 \cdot C) \times D}, \mathbf{E}_{\text{pos}} \in \mathbb{R}^{(N+1) \times D} \quad (1)$$

$$\mathbf{z}'_\ell = \text{MSA}(\text{LN}(\mathbf{z}_{\ell-1})) + \mathbf{z}_{\ell-1}, \quad \ell = 1 \dots L \quad (2)$$

$$\mathbf{z}_\ell = \text{MLP}(\text{LN}(\mathbf{z}'_\ell)) + \mathbf{z}'_\ell, \quad \ell = 1 \dots L \quad (3)$$

$$\mathbf{y} = \text{LN}(\mathbf{z}_L^0) \quad (4)$$

归纳偏置。我们注意到，视觉变换器（Vision Transformer）相比卷积神经网络（CNN）具有更少的图像特定归纳偏置。在CNN中，局部性、二维邻域结构和平移等变性被内置于整个模型的每一层。而在ViT中，只有MLP层是局部且平移等变的，而自注意力层是全局的。二维邻域结构的使用非常有限：仅在模型开始时将图像切割成块，以及在微调时调整不同分辨率图像的位置嵌入（如下所述）。除此之外，初始化时的位置嵌入不包含关于图像块二维位置的信息，所有图像块之间的空间关系都必须从头学习。

混合架构。作为原始图像块的替代方案，输入序列可以由CNN（LeCun等人，1989年）的特征图构成。在此混合模型中，从CNN特征图中提取的块会应用块嵌入投影 $\mathbf{E}$ （公式1）。作为一种特殊情况，这些块的空间尺寸可以为 1×1 ，这意味着输入序列是通过简单展平特征图的空间维度并投影到Transformer维度而获得的。分类输入嵌入和位置嵌入按上述方式添加。

3.2 微调与更高分辨率

通常，我们会在大型数据集上对ViT进行预训练，然后针对（规模较小的）下游任务进行微调。为此，我们会移除预训练时的预测头，并附加一个零初始化的 $D \times K$ 前馈层，其中 K 是下游任务的类别数量。以高于预训练时的分辨率进行微调通常是有益的（Touvron等人，2019；Kolesnikov等人，2020）。当输入更高分辨率的图像时，我们保持图像块大小不变，这会导致有效序列长度增加。Vision Transformer能够处理任意序列长度（受内存限制），但预训练的位置嵌入可能不再具有意义。因此，我们根据预训练位置嵌入在原始图像中的位置，对其进行二维插值。请注意，这种分辨率调整和图像块提取是唯一手动将关于图像二维结构的归纳偏置注入Vision Transformer的环节。

4 实验

我们评估了ResNet、视觉Transformer（ViT）及其混合模型的表征学习能力。为了解各模型的数据需求，我们在不同规模的数据集上进行预训练，并在多个基准任务上评估性能。在考虑模型预训练的计算成本时，ViT表现尤为突出，能以更低的预训练成本在多数识别基准上达到最先进水平。最后，我们通过自监督学习进行了一项小规模实验，结果表明自监督ViT在未来具有广阔的发展前景。

4.1 设置

数据集。为探究模型的可扩展性，我们采用包含1000个类别、130万张图像的ILSVRC-2012 ImageNet数据集（下文简称为ImageNet），其超集ImageNet-21k（Deng等人，2009）包含2.1万个类别、1400万张图像，以及包含1.8万个类别、3.03亿张高分辨率图像的JFT数据集（Sun等人，2017）。我们遵循Kolesnikov等人（2020）的方法，对预训练数据集进行去重处理，排除下游任务测试集中出现的重复数据。将在这些数据集上训练的模型迁移至以下基准任务：使用原始验证标签和经清理的Real标签（Beyer等人，2020）的ImageNet、CIFAR-10/100（Krizhevsky，2009）、Oxford-IIIT Pets（Parkhi等人，2012）以及Oxford Flowers-102（Nilsback & Zisserman，2008）。对于这些数据集，预处理流程均遵循Kolesnikov等人（2020）的方法。

Model	Layers	Hidden size D	MLP size	Heads	Params
ViT-Base	12	768	3072	12	86M
ViT-Large	24	1024	4096	16	307M
ViT-Huge	32	1280	5120	16	632M

Table 1: Details of Vision Transformer model variants.

We also evaluate on the 19-task VTAB classification suite (Zhai et al., 2019b). VTAB evaluates low-data transfer to diverse tasks, using 1 000 training examples per task. The tasks are divided into three groups: *Natural* – tasks like the above, Pets, CIFAR, etc. *Specialized* – medical and satellite imagery, and *Structured* – tasks that require geometric understanding like localization.

Model Variants. We base ViT configurations on those used for BERT (Devlin et al., 2019), as summarized in Table 1. The “Base” and “Large” models are directly adopted from BERT and we add the larger “Huge” model. In what follows we use brief notation to indicate the model size and the input patch size: for instance, ViT-L/16 means the “Large” variant with 16×16 input patch size. Note that the Transformer’s sequence length is inversely proportional to the square of the patch size, thus models with smaller patch size are computationally more expensive.

For the baseline CNNs, we use ResNet (He et al., 2016), but replace the Batch Normalization layers (Ioffe & Szegedy, 2015) with Group Normalization (Wu & He, 2018), and used standardized convolutions (Qiao et al., 2019). These modifications improve transfer (Kolesnikov et al., 2020), and we denote the modified model “ResNet (BiT)”. For the hybrids, we feed the intermediate feature maps into ViT with patch size of one “pixel”. To experiment with different sequence lengths, we either (i) take the output of stage 4 of a regular ResNet50 or (ii) remove stage 4, place the same number of layers in stage 3 (keeping the total number of layers), and take the output of this extended stage 3. Option (ii) results in a 4x longer sequence length, and a more expensive ViT model.

Training & Fine-tuning. We train all models, including ResNets, using Adam (Kingma & Ba, 2015) with $\beta_1 = 0.9$, $\beta_2 = 0.999$, a batch size of 4096 and apply a high weight decay of 0.1, which we found to be useful for transfer of all models (Appendix D.1 shows that, in contrast to common practices, Adam works slightly better than SGD for ResNets in our setting). We use a linear learning rate warmup and decay, see Appendix B.1 for details. For fine-tuning we use SGD with momentum, batch size 512, for all models, see Appendix B.1.1. For ImageNet results in Table 2, we fine-tuned at higher resolution: 512 for ViT-L/16 and 518 for ViT-H/14, and also used Polyak & Juditsky (1992) averaging with a factor of 0.9999 (Ramachandran et al., 2019; Wang et al., 2020b).

Metrics. We report results on downstream datasets either through few-shot or fine-tuning accuracy. Fine-tuning accuracies capture the performance of each model after fine-tuning it on the respective dataset. Few-shot accuracies are obtained by solving a regularized least-squares regression problem that maps the (frozen) representation of a subset of training images to $\{-1, 1\}^K$ target vectors. This formulation allows us to recover the exact solution in closed form. Though we mainly focus on fine-tuning performance, we sometimes use linear few-shot accuracies for fast on-the-fly evaluation where fine-tuning would be too costly.

4.2 COMPARISON TO STATE OF THE ART

We first compare our largest models – ViT-H/14 and ViT-L/16 – to state-of-the-art CNNs from the literature. The first comparison point is Big Transfer (BiT) (Kolesnikov et al., 2020), which performs supervised transfer learning with large ResNets. The second is Noisy Student (Xie et al., 2020), which is a large EfficientNet trained using semi-supervised learning on ImageNet and JFT-300M with the labels removed. Currently, Noisy Student is the state of the art on ImageNet and BiT-L on the other datasets reported here. All models were trained on TPUv3 hardware, and we report the number of TPUv3-core-days taken to pre-train each of them, that is, the number of TPU v3 cores (2 per chip) used for training multiplied by the training time in days.

Table 2 shows the results. The smaller ViT-L/16 model pre-trained on JFT-300M outperforms BiT-L (which is pre-trained on the same dataset) on all tasks, while requiring substantially less computational resources to train. The larger model, ViT-H/14, further improves the performance, especially on the more challenging datasets – ImageNet, CIFAR-100, and the VTAB suite. Interestingly, this

Model	Layers	Hidden size D	MLP size	Heads	Params
ViT-Base	12	768	3072	12	86M
ViT-Large	24	1024	4096	16	307M
ViT-Huge	32	1280	5120	16	632M

表1: Vision Transformer模型变体详情。

我们还在包含19个任务的VTAB分类套件 (Zhai等人, 2019b) 上进行了评估。VTAB通过每个任务使用1000个训练样本来评估对多样化任务的低数据迁移能力。这些任务分为三组: *Natural*——类似上述任务、Pets、CIFAR等; *Specialized*——医学与卫星图像任务; 以及 *Structured*——需要几何理解 (如定位) 的任务。

模型变体。我们基于BERT (Devlin等人, 2019) 所使用的配置来设定ViT的配置, 如表1所示。"Base"和"Large"模型直接沿用自BERT, 我们还增加了更大的"Huge"模型。在下文中, 我们使用简略表示法来指代模型尺寸和输入图像块尺寸: 例如, ViT-L/16表示采用 16×16 输入图像块尺寸的"Large"变体。请注意, Transformer的序列长度与图像块尺寸的平方成反比, 因此图像块尺寸较小的模型在计算上更为昂贵。

对于基线CNN, 我们采用ResNet (He等人, 2016), 但将批量归一化层 (Ioffe & Szegedy, 2015) 替换为组归一化 (Wu & He, 2018), 并使用标准化卷积 (Qiao等人, 2019)。这些改进提升了迁移性能 (Kolesnikov等人, 2020), 我们将改进后的模型称为"ResNet (BiT)"。对于混合模型, 我们将中间特征图输入到ViT中, 其补丁大小为单个"像素"。为了尝试不同的序列长度, 我们选择两种方案: (i) 采用常规ResNet50第4阶段的输出, 或(ii) 移除第4阶段, 将相同数量的层置于第3阶段 (保持总层数不变), 并采用扩展后第3阶段的输出。方案(ii) 会产生4倍长的序列长度, 并构建计算成本更高的ViT模型。

训练与微调。我们使用Adam优化器 (Kingma & Ba, 2015) 训练所有模型 (包括ResNets), 参数设置为 $\beta_1=0.9$ 、 $\beta_2=0.999$, 批大小为4096, 并采用0.1的高权重衰减——我们发现这对所有模型的迁移都有帮助 (附录D.1显示, 与常见做法不同, 在我们的设定中Adam对ResNets的效果略优于SGD)。我们采用线性学习率预热与衰减策略, 详见附录B.1。对于微调阶段, 所有模型均使用带动量的SGD优化器, 批大小为512, 具体见附录B.1.1。针对表2中的ImageNet结果, 我们以更高分辨率进行微调: ViT-L/16为512, ViT-H/14为518, 同时采用Polyak & Juditsky (1992) 平均法, 衰减因子为0.9999 (Ramachandran等, 2019; Wang等, 2020b)。

指标。我们通过少样本或微调准确率报告下游数据集的结果。微调准确率反映了各模型在相应数据集上微调后的性能。少样本准确率通过求解正则化最小二乘回归问题获得, 该问题将训练图像子集的 (冻结) 表示映射到 $\{v^*\}$ 目标向量。这一表述使我们能够以闭式解的形式得到精确解。尽管我们主要关注微调性能, 但在微调成本过高时, 我们有时会使用线性少样本准确率进行快速即时评估。

4.2 与现有技术的比较

我们首先将我们最大的模型——ViT-H/14和ViT-L/16——与文献中最先进的CNN进行比较。第一个比较点是Big Transfer (BiT) (Kolesnikov等人, 2020年), 它使用大型ResNet进行监督式迁移学习。第二个是Noisy Student (Xie等人, 2020年), 这是一个在ImageNet和JFT-300M上使用半监督学习 (标签已移除) 训练的大型EfficientNet。目前, Noisy Student在ImageNet上处于领先地位, 而BiT-L在本文报告的其他数据集上表现最佳。所有模型均在TPUv3硬件上训练, 我们报告了预训练每个模型所消耗的TPUv3核心天数, 即用于训练的TPU v3核心数 (每个芯片2个核心) 乘以训练天数。

表2展示了结果。在JFT-300M数据集上预训练的较小模型ViT-L/16在所有任务上都优于同数据集预训练的BiT-L, 同时训练所需的计算资源显著减少。更大的ViT-H/14模型则进一步提升了性能, 尤其在更具挑战性的数据集——ImageNet、CIFAR-100和VTAB套件上表现突出。有趣的是, 这一

	Ours-JFT (ViT-H/14)	Ours-JFT (ViT-L/16)	Ours-I21k (ViT-L/16)	BiT-L (ResNet152x4)	Noisy Student (EfficientNet-L2)
ImageNet	88.55 ± 0.04	87.76 ± 0.03	85.30 ± 0.02	87.54 ± 0.02	88.4/88.5*
ImageNet ReaL	90.72 ± 0.05	90.54 ± 0.03	88.62 ± 0.05	90.54	90.55
CIFAR-10	99.50 ± 0.06	99.42 ± 0.03	99.15 ± 0.03	99.37 ± 0.06	—
CIFAR-100	94.55 ± 0.04	93.90 ± 0.05	93.25 ± 0.05	93.51 ± 0.08	—
Oxford-IIIT Pets	97.56 ± 0.03	97.32 ± 0.11	94.67 ± 0.15	96.62 ± 0.23	—
Oxford Flowers-102	99.68 ± 0.02	99.74 ± 0.00	99.61 ± 0.02	99.63 ± 0.03	—
VTAB (19 tasks)	77.63 ± 0.23	76.28 ± 0.46	72.72 ± 0.21	76.29 ± 1.70	—
TPUv3-core-days	2.5k	0.68k	0.23k	9.9k	12.3k

Table 2: Comparison with state of the art on popular image classification benchmarks. We report mean and standard deviation of the accuracies, averaged over three fine-tuning runs. Vision Transformer models pre-trained on the JFT-300M dataset outperform ResNet-based baselines on all datasets, while taking substantially less computational resources to pre-train. ViT pre-trained on the smaller public ImageNet-21k dataset performs well too. *Slightly improved 88.5% result reported in Touvron et al. (2020).

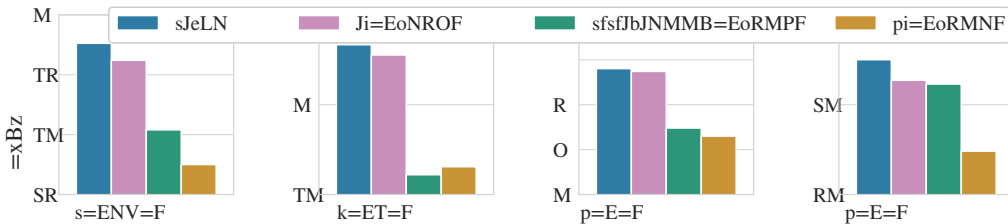


Figure 2: Breakdown of VTAB performance in *Natural*, *Specialized*, and *Structured* task groups.

model still took substantially less compute to pre-train than prior state of the art. However, we note that pre-training efficiency may be affected not only by the architecture choice, but also other parameters, such as training schedule, optimizer, weight decay, etc. We provide a controlled study of performance vs. compute for different architectures in Section 4.4. Finally, the ViT-L/16 model pre-trained on the public ImageNet-21k dataset performs well on most datasets too, while taking fewer resources to pre-train: it could be trained using a standard cloud TPUv3 with 8 cores in approximately 30 days.

Figure 2 decomposes the VTAB tasks into their respective groups, and compares to previous SOTA methods on this benchmark: BiT, VIVI – a ResNet co-trained on ImageNet and Youtube (Tschannen et al., 2020), and S4L – supervised plus semi-supervised learning on ImageNet (Zhai et al., 2019a). ViT-H/14 outperforms BiT-R152x4, and other methods, on the *Natural* and *Structured* tasks. On the *Specialized* the performance of the top two models is similar.

4.3 PRE-TRAINING DATA REQUIREMENTS

The Vision Transformer performs well when pre-trained on a large JFT-300M dataset. With fewer inductive biases for vision than ResNets, how crucial is the dataset size? We perform two series of experiments.

First, we pre-train ViT models on datasets of increasing size: ImageNet, ImageNet-21k, and JFT-300M. To boost the performance on the smaller datasets, we optimize three basic regularization parameters – weight decay, dropout, and label smoothing. Figure 3 shows the results after fine-tuning to ImageNet (results on other datasets are shown in Table 5)². When pre-trained on the smallest dataset, ImageNet, ViT-Large models underperform compared to ViT-Base models, despite (moderate) regularization. With ImageNet-21k pre-training, their performances are similar. Only with JFT-300M, do we see the full benefit of larger models. Figure 3 also shows the performance

²Note that the ImageNet pre-trained models are also fine-tuned, but again on ImageNet. This is because the resolution increase during fine-tuning improves the performance.

	Ours-JFT (ViT-H/14)	Ours-JFT (ViT-L/16)	Ours-I21k (ViT-L/16)	BiT-L (ResNet152x4)	Noisy Student (EfficientNet-L2)
ImageNet	88.55 ± 0.04	87.76 ± 0.03	85.30 ± 0.02	87.54 ± 0.02	88.4/88.5*
ImageNet ReaL	90.72 ± 0.05	90.54 ± 0.03	88.62 ± 0.05	90.54	90.55
CIFAR-10	99.50 ± 0.06	99.42 ± 0.03	99.15 ± 0.03	99.37 ± 0.06	—
CIFAR-100	94.55 ± 0.04	93.90 ± 0.05	93.25 ± 0.05	93.51 ± 0.08	—
Oxford-IIIT Pets	97.56 ± 0.03	97.32 ± 0.11	94.67 ± 0.15	96.62 ± 0.23	—
Oxford Flowers-102	99.68 ± 0.02	99.74 ± 0.00	99.61 ± 0.02	99.63 ± 0.03	—
VTAB (19 tasks)	77.63 ± 0.23	76.28 ± 0.46	72.72 ± 0.21	76.29 ± 1.70	—
TPUv3-core-days	2.5k	0.68k	0.23k	9.9k	12.3k

表2: 在主流图像分类基准上与最先进技术比较。我们报告了三次微调运行平均后的准确率均值及标准差。在JFT-300M数据集上预训练的Vision Transformer模型在所有数据集上均优于基于ResNet的基线模型, 且预训练所需的计算资源显著减少。在较小的公开数据集ImageNet-21k上预训练的ViT模型同样表现优异。*Touvron等人(2020)中报告的88.5%结果略有提升。

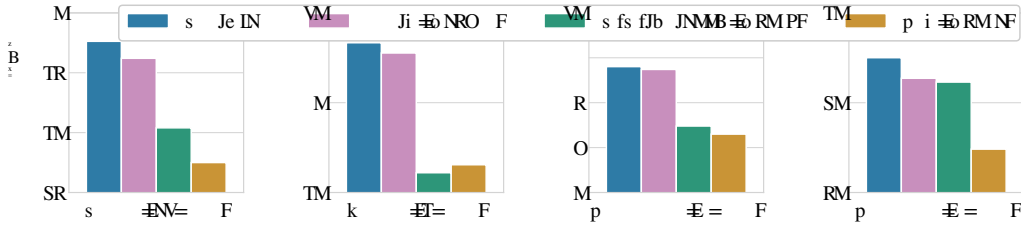


图2: 细分 VTAB在Natural、Specialized和Structured中的性能表现 ed 任务组。

该模型的预训练计算量仍远低于之前的先进技术。然而, 我们注意到预训练效率不仅受架构选择的影响, 还可能受到其他参数(如训练计划、优化器、权重衰减等)的影响。我们在第4.4节中对不同架构的性能与计算量进行了对照研究。最后, 基于公开数据集ImageNet-21k预训练的ViT-L/16模型在多数数据集上也表现优异, 且所需预训练资源更少: 使用配备8核的标准云端TPUv3约30天即可完成训练。

图2将VTAB任务分解为各自的组别, 并与该基准测试中先前的最先进方法进行比较: BiT、ViViT——一种在ImageNet和YouTube上联合训练的ResNet (Tschannen等人, 2020年), 以及S4L——在ImageNet上的监督加半监督学习 (Zhai等人, 2019a)。ViT-H/14在Natural和Structured任务上优于BiT-R152x4及其他方法。在Specialized任务上, 两种顶级模型的性能相近。

4.3 预训练数据要求

视觉Transformer在大型JFT-300M数据集上进行预训练时表现优异。相比ResNets, 其视觉归纳偏置更少, 那么数据集规模究竟有多关键? 我们进行了两个系列的实验。

首先, 我们在规模递增的数据集上预训练ViT模型: ImageNet、ImageNet-21k和JFT-300M。为提升在较小数据集上的性能, 我们优化了三个基础正则化参数——权重衰减、dropout和标签平滑。图3展示了在ImageNet上微调后的结果(其他数据集的结果见表5)²。当在最小的数据集ImageNet上预训练时, 即使采用了(适度的)正则化, ViT-Large模型的表现仍不及ViT-Base模型。使用ImageNet-21k预训练后, 两者的性能趋于相近。只有在JFT-300M上预训练时, 我们才能看到更大模型的全部优势。图3还展示了性能

²Note that the ImageNet pre-trained models are also fine-tuned, but again on ImageNet. This is because the resolution increase during fine-tuning improves the performance.

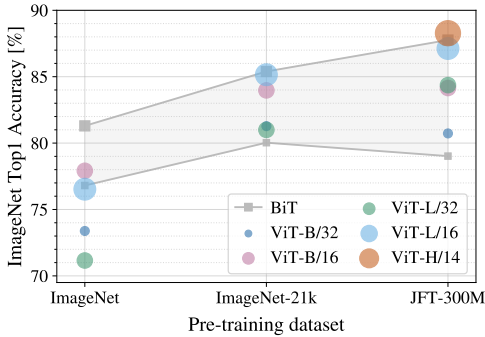


Figure 3: Transfer to ImageNet. While large ViT models perform worse than BiT ResNets (shaded area) when pre-trained on small datasets, they shine when pre-trained on larger datasets. Similarly, larger ViT variants overtake smaller ones as the dataset grows.

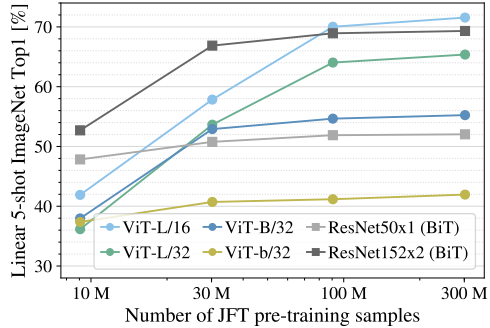


Figure 4: Linear few-shot evaluation on ImageNet versus pre-training size. ResNets perform better with smaller pre-training datasets but plateau sooner than ViT, which performs better with larger pre-training. ViT-b is ViT-B with all hidden dimensions halved.

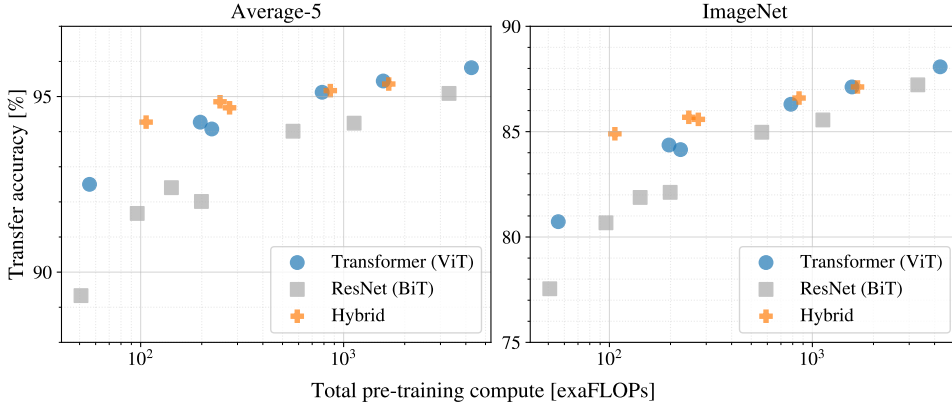


Figure 5: Performance versus pre-training compute for different architectures: Vision Transformers, ResNets, and hybrids. Vision Transformers generally outperform ResNets with the same computational budget. Hybrids improve upon pure Transformers for smaller model sizes, but the gap vanishes for larger models.

region spanned by BiT models of different sizes. The BiT CNNs outperform ViT on ImageNet, but with the larger datasets, ViT overtakes.

Second, we train our models on random subsets of 9M, 30M, and 90M as well as the full JFT-300M dataset. We do not perform additional regularization on the smaller subsets and use the same hyper-parameters for all settings. This way, we assess the intrinsic model properties, and not the effect of regularization. We do, however, use early-stopping, and report the best validation accuracy achieved during training. To save compute, we report few-shot linear accuracy instead of full fine-tuning accuracy. Figure 4 contains the results. Vision Transformers overfit more than ResNets with comparable computational cost on smaller datasets. For example, ViT-B/32 is slightly faster than ResNet50; it performs much worse on the 9M subset, but better on 90M+ subsets. The same is true for ResNet152x2 and ViT-L/16. This result reinforces the intuition that the convolutional inductive bias is useful for smaller datasets, but for larger ones, learning the relevant patterns directly from data is sufficient, even beneficial.

Overall, the few-shot results on ImageNet (Figure 4), as well as the low-data results on VTAB (Table 2) seem promising for very low-data transfer. Further analysis of few-shot properties of ViT is an exciting direction of future work.

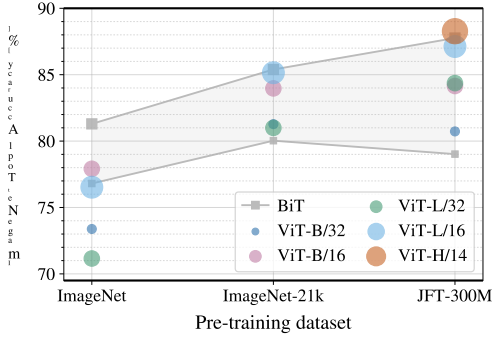


图3: 向ImageNet的迁移。当在小型数据集上进行预训练时, 大型ViT模型的表现不如BiT ResNets (阴影区域), 但在更大数据集上预训练时则表现出色。同样地, 随着数据集的增大, 较大的ViT变体会超越较小的变体。

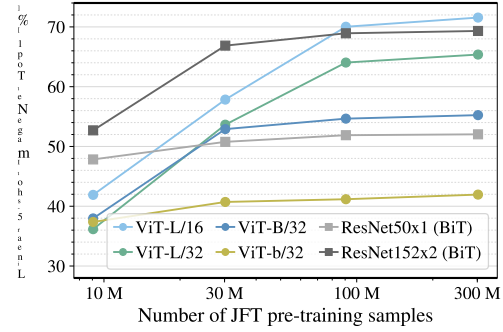


图4: ImageNet上的线性少样本评估与预训练规模的关系。ResNet在较小的预训练数据集上表现更好, 但比ViT更早进入平台期; ViT在更大的预训练规模下表现更优。ViT-b是将所有隐藏维度减半的ViT-B模型。

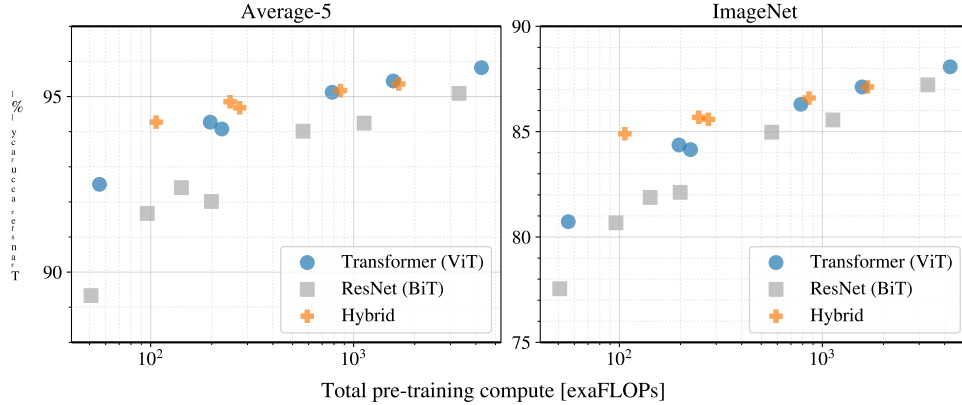


图5: 不同架构的性能与预训练计算量的关系: 视觉Transformer、ResNet及混合架构。在相同计算预算下, 视觉Transformer通常优于ResNet。对于较小规模的模型, 混合架构较纯Transformer有所改进, 但随着模型规模增大, 这种差距逐渐消失。

不同尺寸BiT模型所覆盖的区域。BiT CNN在ImageNet上表现优于ViT, 但随着数据集规模扩大, ViT实现了反超。

其次, 我们在9M、30M和90M的随机子集以及完整的JFT-300M数据集上训练模型。对于较小的子集, 我们没有进行额外的正则化处理, 并且所有设置都使用相同的超参数。通过这种方式, 我们评估模型的内在特性, 而不是正则化的效果。然而, 我们确实使用了早停法, 并报告了训练过程中达到的最佳验证准确率。为了节省计算资源, 我们报告的是少样本线性准确率, 而不是完整的微调准确率。图4展示了结果。在计算成本相当的情况下, 视觉变换器在较小数据集上的过拟合程度高于ResNet。例如, ViT-B/32比ResNet50稍快; 它在9M子集上表现差得多, 但在90M+子集上表现更好。ResNet152x2和ViT-L/16也有类似情况。这一结果进一步印证了直觉: 卷积归纳偏置对于较小数据集是有用的, 但对于更大的数据集, 直接从数据中学习相关模式就足够了, 甚至更有益处。

总体而言, ImageNet上的少样本学习结果 (图4) 以及VTAB上的低数据量结果 (表2) 对于极低数据量迁移任务展现出良好前景。进一步分析ViT的少样本特性是未来工作中值得探索的方向。

4.4 SCALING STUDY

We perform a controlled scaling study of different models by evaluating transfer performance from JFT-300M. In this setting data size does not bottleneck the models’ performances, and we assess performance versus pre-training cost of each model. The model set includes: 7 ResNets, R50x1, R50x2 R101x1, R152x1, R152x2, pre-trained for 7 epochs, plus R152x2 and R200x3 pre-trained for 14 epochs; 6 Vision Transformers, ViT-B/32, B/16, L/32, L/16, pre-trained for 7 epochs, plus L/16 and H/14 pre-trained for 14 epochs; and 5 hybrids, R50+ViT-B/32, B/16, L/32, L/16 pre-trained for 7 epochs, plus R50+ViT-L/16 pre-trained for 14 epochs (for hybrids, the number at the end of the model name stands not for the patch size, but for the total dowsampling ratio in the ResNet backbone).

Figure 5 contains the transfer performance versus total pre-training compute (see Appendix D.5 for details on computational costs). Detailed results per model are provided in Table 6 in the Appendix. A few patterns can be observed. First, Vision Transformers dominate ResNets on the performance/compute trade-off. ViT uses approximately $2 - 4\times$ less compute to attain the same performance (average over 5 datasets). Second, hybrids slightly outperform ViT at small computational budgets, but the difference vanishes for larger models. This result is somewhat surprising, since one might expect convolutional local feature processing to assist ViT at any size. Third, Vision Transformers appear not to saturate within the range tried, motivating future scaling efforts.

4.5 INSPECTING VISION TRANSFORMER

To begin to understand how the Vision Transformer processes image data, we analyze its internal representations. The first layer of the Vision Transformer linearly projects the flattened patches into a lower-dimensional space (Eq. 1). Figure 7 (left) shows the top principal components of the learned embedding filters. The components resemble plausible basis functions for a low-dimensional representation of the fine structure within each patch.

After the projection, a learned position embedding is added to the patch representations. Figure 7 (center) shows that the model learns to encode distance within the image in the similarity of position embeddings, i.e. closer patches tend to have more similar position embeddings. Further, the row-column structure appears; patches in the same row/column have similar embeddings. Finally, a sinusoidal structure is sometimes apparent for larger grids (Appendix D). That the position embeddings learn to represent 2D image topology explains why hand-crafted 2D-aware embedding variants do not yield improvements (Appendix D.4).

Self-attention allows ViT to integrate information across the entire image even in the lowest layers. We investigate to what degree the network makes use of this capability. Specifically, we compute the average distance in image space across which information is integrated, based on the attention weights (Figure 7, right). This “attention distance” is analogous to receptive field size in CNNs.

We find that some heads attend to most of the image already in the lowest layers, showing that the ability to integrate information globally is indeed used by the model. Other attention heads have consistently small attention distances in the low layers. This highly localized attention is less pronounced in hybrid models that apply a ResNet before the Transformer (Figure 7, right), suggesting that it may serve a similar function as early convolutional layers in CNNs. Further, the attention distance increases with network depth. Globally, we find that the model attends to image regions that are semantically relevant for classification (Figure 6).

4.6 SELF-SUPERVISION

Transformers show impressive performance on NLP tasks. However, much of their success stems not only from their excellent scalability but also from large scale self-supervised pre-training (Devlin

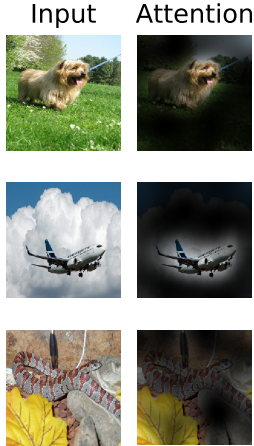


Figure 6: Representative examples of attention from the output token to the input space. See Appendix D.7 for details.

4.4 缩放研究

我们通过对从JFT-300M迁移性能的评估，对不同模型进行了受控的缩放研究。在此设定下，数据规模不会成为模型性能的瓶颈，我们评估了各模型性能与预训练成本的关系。模型集合包括：7个ResNet模型（R50x1、R50x2、R101x1、R152x1、R152x2预训练7个周期，以及R152x2和R200x3预训练14个周期）；6个视觉Transformer模型（ViT-B/32、B/16、L/32、L/16预训练7个周期，以及L/16和H/14预训练14个周期）；以及5个混合模型（R50+ViT-B/32、B/16、L/32、L/16预训练7个周期，外加R50+ViT-L/16预训练14个周期）。对于混合模型，其名称末尾的数字不代表图像块大小，而是指ResNet骨干网络中的总下采样率。

图5展示了迁移性能与总预训练计算量的关系（计算成本详情见附录D.5）。各模型详细结果见附录表6。我们可以观察到几个现象：首先，在性能与计算量的权衡中，视觉Transformer全面优于ResNet。ViT达到相同性能（5个数据集平均值）所需计算量约为ResNet的2–4 $\times$ 分之一。其次，在较小计算预算下，混合模型略优于ViT，但随着模型规模增大，差异逐渐消失。这一结果有些出人意料，因为通常认为卷积局部特征处理能力应能辅助任何规模的ViT。第三，视觉Transformer在实验规模范围内未出现性能饱和，这为未来的扩展研究提供了动力。

4.5 视觉变换器检查

为了理解Vision Transformer如何处理图像数据，我们分析了其内部表征。Vision Transformer的第一层将展平的图像块线性投影到低维空间（公式1）。图7（左）展示了所学嵌入滤波器的主要主成分。这些成分类似于每个图像块内部精细结构的低维表示中合理的基函数。

在投影之后，学习得到的位置嵌入被添加到图像块表示中。图7（中心）显示，模型学会了在位置嵌入的相似性中对图像内的距离进行编码，即更接近的图像块往往具有更相似的位置嵌入。此外，行-列结构也显现出来；同一行/列的图像块具有相似的嵌入。最后，对于较大的网格，有时会出现正弦结构（附录D）。位置嵌入学会表示二维图像拓扑，这解释了为什么手工设计的具有二维感知的嵌入变体没有带来改进（附录D.4）。

自注意力机制使得ViT即使在最底层也能整合整个图像的信息。我们探究了网络在多大程度上利用了这种能力。具体而言，我们基于注意力权重（图7右）计算了图像空间中信息被整合的平均距离。这种“注意力距离”类似于CNN中的感受野大小。

我们发现，一些注意力头在最低层就已经关注到图像的大部分区域，这表明模型确实使用了全局整合信息的能力。其他注意力头在低层始终保持着较小的注意力距离。这种高度局部化的注意力在Transformer之前应用ResNet的混合模型中不太明显（图7，右），这表明它可能类似于CNN中早期卷积层的功能。此外，注意力距离随着网络深度的增加而增加。总体而言，我们发现模型关注的图像区域在语义上与分类相关（图6）。

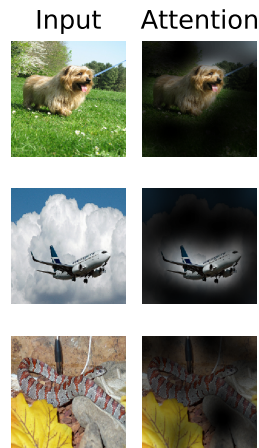


图6：输出标记对输入空间的注意力代表性示例。详情见附录D.7。

4.6 自监督学习

Transformer在自然语言处理任务上表现出令人印象深刻的性能。然而，其成功很大程度上不仅源于出色的可扩展性，还来自大规模自监督预训练（Devlin

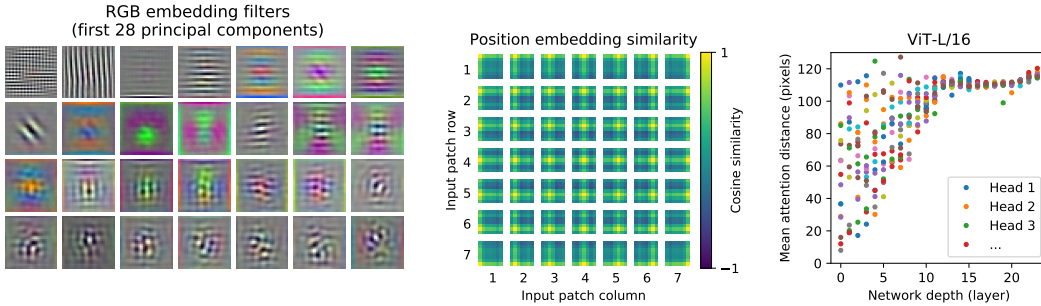


Figure 7: **Left:** Filters of the initial linear embedding of RGB values of ViT-L/32. **Center:** Similarity of position embeddings of ViT-L/32. Tiles show the cosine similarity between the position embedding of the patch with the indicated row and column and the position embeddings of all other patches. **Right:** Size of attended area by head and network depth. Each dot shows the mean attention distance across images for one of 16 heads at one layer. See Appendix D.7 for details.

et al., 2019; Radford et al., 2018). We also perform a preliminary exploration on *masked patch prediction* for self-supervision, mimicking the masked language modeling task used in BERT. With self-supervised pre-training, our smaller ViT-B/16 model achieves 79.9% accuracy on ImageNet, a significant improvement of 2% to training from scratch, but still 4% behind supervised pre-training. Appendix B.1.2 contains further details. We leave exploration of contrastive pre-training (Chen et al., 2020b; He et al., 2020; Bachman et al., 2019; Hénaff et al., 2020) to future work.

5 CONCLUSION

We have explored the direct application of Transformers to image recognition. Unlike prior works using self-attention in computer vision, we do not introduce image-specific inductive biases into the architecture apart from the initial patch extraction step. Instead, we interpret an image as a sequence of patches and process it by a standard Transformer encoder as used in NLP. This simple, yet scalable, strategy works surprisingly well when coupled with pre-training on large datasets. Thus, Vision Transformer matches or exceeds the state of the art on many image classification datasets, whilst being relatively cheap to pre-train.

While these initial results are encouraging, many challenges remain. One is to apply ViT to other computer vision tasks, such as detection and segmentation. Our results, coupled with those in Carion et al. (2020), indicate the promise of this approach. Another challenge is to continue exploring self-supervised pre-training methods. Our initial experiments show improvement from self-supervised pre-training, but there is still large gap between self-supervised and large-scale supervised pre-training. Finally, further scaling of ViT would likely lead to improved performance.

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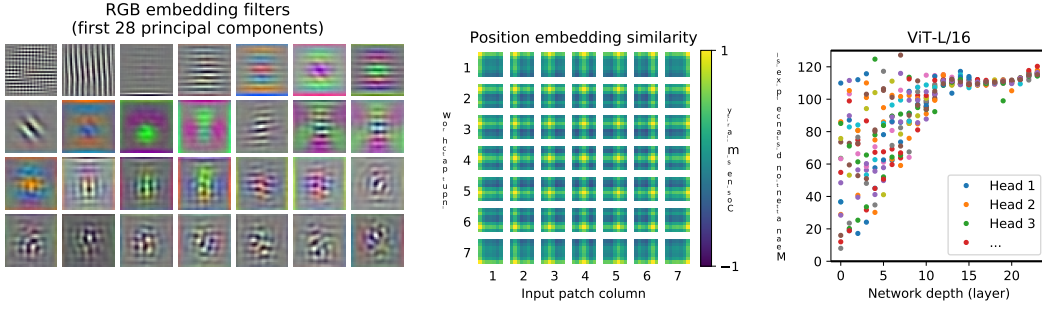


图7: 左: ViT-L/32的RGB值初始线性嵌入的滤波器。中: ViT-L/32位置嵌入的相似性。每个图块显示指定行和列的补丁位置嵌入与所有其他补丁位置嵌入之间的余弦相似度。右: 按注意力头和网络深度划分的关注区域大小。每个点表示某一层中16个注意力头之一在图像上的平均注意力距离。详见附录D.7。

等人, 2019; Radford等人, 2018)。我们还对*masked patch prediction*进行了初步探索, 用于自监督学习, 模仿了BERT中使用的掩码语言建模任务。通过自监督预训练, 我们较小的ViT-B/16模型在ImageNet上达到了79.9%的准确率, 相比从头训练显著提升了2%, 但仍比有监督预训练低4%。附录B.1.2包含了更多细节。我们将对比式预训练(Chen等人, 2020b; He等人, 2020; Bachman等人, 2019; Hénaff等人, 2020)的探索留待未来工作。

5 结论

我们探索了将Transformer直接应用于图像识别的方法。与先前在计算机视觉中使用自注意力的工作不同, 除了初始的图像块提取步骤外, 我们并未在架构中引入针对图像的特定归纳偏置。相反, 我们将图像解释为一系列图像块, 并使用自然语言处理中常用的标准Transformer编码器进行处理。这种简单而可扩展的策略, 在结合大型数据集预训练时表现出惊人的效果。因此, Vision Transformer在许多图像分类数据集上达到或超越了现有技术水平, 同时预训练成本相对较低。

尽管这些初步结果令人鼓舞, 但许多挑战依然存在。其一是将ViT应用于其他计算机视觉任务, 如检测与分割。我们的研究结果与Carion等人(2020)的工作共同印证了该方法的潜力。另一挑战是持续探索自监督预训练方法。我们的初步实验显示自监督预训练已带来改进, 但其与大规模监督预训练之间仍存在显著差距。最后, 进一步扩大ViT的规模很可能会带来性能提升。

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Models	Dataset	Epochs	Base LR	LR decay	Weight decay	Dropout
ViT-B/{16,32}	JFT-300M	7	$8 \cdot 10^{-4}$	linear	0.1	0.0
ViT-L/32	JFT-300M	7	$6 \cdot 10^{-4}$	linear	0.1	0.0
ViT-L/16	JFT-300M	7/14	$4 \cdot 10^{-4}$	linear	0.1	0.0
ViT-H/14	JFT-300M	14	$3 \cdot 10^{-4}$	linear	0.1	0.0
R50x{1,2}	JFT-300M	7	10^{-3}	linear	0.1	0.0
R101x1	JFT-300M	7	$8 \cdot 10^{-4}$	linear	0.1	0.0
R152x{1,2}	JFT-300M	7	$6 \cdot 10^{-4}$	linear	0.1	0.0
R50+ViT-B/{16,32}	JFT-300M	7	$8 \cdot 10^{-4}$	linear	0.1	0.0
R50+ViT-L/32	JFT-300M	7	$2 \cdot 10^{-4}$	linear	0.1	0.0
R50+ViT-L/16	JFT-300M	7/14	$4 \cdot 10^{-4}$	linear	0.1	0.0
ViT-B/{16,32}	ImageNet-21k	90	10^{-3}	linear	0.03	0.1
ViT-L/{16,32}	ImageNet-21k	30/90	10^{-3}	linear	0.03	0.1
ViT-*	ImageNet	300	$3 \cdot 10^{-3}$	cosine	0.3	0.1

Table 3: Hyperparameters for training. All models are trained with a batch size of 4096 and learning rate warmup of 10k steps. For ImageNet we found it beneficial to additionally apply gradient clipping at global norm 1. Training resolution is 224.

APPENDIX

A MULTIHEAD SELF-ATTENTION

Standard $\mathbf{qkv}$ self-attention (SA, Vaswani et al. (2017)) is a popular building block for neural architectures. For each element in an input sequence $\mathbf{z} \in \mathbb{R}^{N \times D}$, we compute a weighted sum over all values $\mathbf{v}$ in the sequence. The attention weights A_{ij} are based on the pairwise similarity between two elements of the sequence and their respective query $\mathbf{q}^i$ and key $\mathbf{k}^j$ representations.

$$[\mathbf{q}, \mathbf{k}, \mathbf{v}] = \mathbf{z} \mathbf{U}_{qkv} \quad \mathbf{U}_{qkv} \in \mathbb{R}^{D \times 3D_h}, \quad (5)$$

$$A = \text{softmax} \left(\frac{\mathbf{qk}^\top}{\sqrt{D_h}} \right) \quad A \in \mathbb{R}^{N \times N}, \quad (6)$$

$$\text{SA}(\mathbf{z}) = A \mathbf{v}. \quad (7)$$

Multihead self-attention (MSA) is an extension of SA in which we run k self-attention operations, called “heads”, in parallel, and project their concatenated outputs. To keep compute and number of parameters constant when changing k , D_h (Eq. 5) is typically set to D/k .

$$\text{MSA}(\mathbf{z}) = [\text{SA}_1(\mathbf{z}); \text{SA}_2(\mathbf{z}); \dots; \text{SA}_k(\mathbf{z})] \mathbf{U}_{msa} \quad \mathbf{U}_{msa} \in \mathbb{R}^{k \cdot D_h \times D} \quad (8)$$

B EXPERIMENT DETAILS

B.1 TRAINING

Table 3 summarizes our training setups for our different models. We found strong regularization to be key when training models from scratch on ImageNet. Dropout, when used, is applied after every dense layer except for the the $\mathbf{qkv}$ -projections and directly after adding positional- to patch embeddings. Hybrid models are trained with the exact setup as their ViT counterparts. Finally, all training is done on resolution 224.

B.1.1 FINE-TUNING

We fine-tune all ViT models using SGD with a momentum of 0.9. We run a small grid search over learning rates, see learning rate ranges in Table 4. To do so, we use small sub-splits from the training set (10% for Pets and Flowers, 2% for CIFAR, 1% ImageNet) as development set and train on the remaining data. For final results we train on the entire training set and evaluate on the respective test data. For fine-tuning ResNets and hybrid models we use the exact same setup, with the only exception of ImageNet where we add another value 0.06 to the learning rate sweep. Additionally,

Models	Dataset	Epochs	Base LR	LR decay	Weight decay	Dropout
ViT-B/{16,32}	JFT-300M	7	$8 \cdot 10^{-4}$	linear	0.1	0.0
ViT-L/32	JFT-300M	7	$6 \cdot 10^{-4}$	linear	0.1	0.0
ViT-L/16	JFT-300M	7/14	$4 \cdot 10^{-4}$	linear	0.1	0.0
ViT-H/14	JFT-300M	14	$3 \cdot 10^{-4}$	linear	0.1	0.0
R50x{1,2}	JFT-300M	7	10^{-3}	linear	0.1	0.0
R101x1	JFT-300M	7	$8 \cdot 10^{-4}$	linear	0.1	0.0
R152x{1,2}	JFT-300M	7	$6 \cdot 10^{-4}$	linear	0.1	0.0
R50+ViT-B/{16,32}	JFT-300M	7	$8 \cdot 10^{-4}$	linear	0.1	0.0
R50+ViT-L/32	JFT-300M	7	$2 \cdot 10^{-4}$	linear	0.1	0.0
R50+ViT-L/16	JFT-300M	7/14	$4 \cdot 10^{-4}$	linear	0.1	0.0
ViT-B/{16,32}	ImageNet-21k	90	10^{-3}	linear	0.03	0.1
ViT-L/{16,32}	ImageNet-21k	30/90	10^{-3}	linear	0.03	0.1
ViT-*	ImageNet	300	$3 \cdot 10^{-3}$	cosine	0.3	0.1

表3: 训练超参数。所有模型均以4096的批次大小和10k步的学习率预热进行训练。对于ImageNet, 我们发现额外应用全局范数为1的梯度裁剪是有益的。训练分辨率为224。

附录

多头自注意力

标准 **qkv** 自注意力机制 (SA, Vaswani 等人 (2017)) 是神经架构中常用的基础模块。对于输入序列 $\mathbf{z} \in \mathbb{R}^{N \times D}$ 中的每个元素, 我们计算序列中所有值 $\mathbf{v}$ 的加权和。注意力权重 A_{ij} 基于序列中两个元素之间的成对相似度及其各自的查询 $\mathbf{q}^i$ 与键 $\mathbf{k}^j$ 表示。

$$[\mathbf{q}, \mathbf{k}, \mathbf{v}] = \mathbf{z} \mathbf{U}_{qkv} \quad \mathbf{U}_{qkv} \in \mathbb{R}^{D \times 3D_h}, \quad (5)$$

$$A = \text{softmax} \left(\frac{\mathbf{q} \mathbf{k}^\top}{\sqrt{D_h}} \right) \quad A \in \mathbb{R}^{N \times N}, \quad (6)$$

$$\text{SA}(\mathbf{z}) = A \mathbf{v}. \quad (7)$$

多头自注意力 (MSA) 是自注意力 (SA) 的扩展, 其中我们并行运行 k 个称为“头”的自注意力操作, 并将它们的拼接输出进行投影。为了在改变 k 时保持计算量和参数数量不变, D_h (式5) 通常被设定为 D/k 。

$$\text{MSA}(\mathbf{z}) = [\text{SA}_1(\mathbf{z}); \text{SA}_2(\mathbf{z}); \dots; \text{SA}_k(\mathbf{z})] \mathbf{U}_{msa} \quad \mathbf{U}_{msa} \in \mathbb{R}^{k \cdot D_h \times D} \quad (8)$$

B 实验详情

B.1 训练

表3总结了针对不同模型的训练设置。我们发现在ImageNet上从头训练模型时, 强正则化是关键。当使用Dropout时, 除qkv投影层外, 每个稠密层后都会应用, 并在位置嵌入与图像块嵌入相加后直接使用。混合模型的训练设置与其对应的ViT模型完全相同。最后, 所有训练均在224分辨率下完成。

B.1.1 微调

我们使用动量为0.9的SGD对所有ViT模型进行微调。我们对学习率进行了小范围的网格搜索, 具体范围见表4。为此, 我们从训练集中划分出小部分子集 (Pets和Flowers数据集取10%, CIFAR取2%, ImageNet取1%) 作为开发集, 并在剩余数据上进行训练。最终结果采用完整训练集进行训练, 并在相应测试数据上评估。对于ResNets和混合模型微调, 我们采用完全相同的设置, 仅在ImageNet数据集上额外增加了0.06这一学习率搜索值。此外,

Dataset	Steps	Base LR
ImageNet	20 000	{0.003, 0.01, 0.03, 0.06}
CIFAR100	10 000	{0.001, 0.003, 0.01, 0.03}
CIFAR10	10 000	{0.001, 0.003, 0.01, 0.03}
Oxford-IIIT Pets	500	{0.001, 0.003, 0.01, 0.03}
Oxford Flowers-102	500	{0.001, 0.003, 0.01, 0.03}
VTAB (19 tasks)	2 500	0.01

Table 4: Hyperparameters for fine-tuning. All models are fine-tuned with cosine learning rate decay, a batch size of 512, no weight decay, and grad clipping at global norm 1. If not mentioned otherwise, fine-tuning resolution is 384.

for ResNets we also run the setup of Kolesnikov et al. (2020) and select the best results across this run and our sweep. Finally, if not mentioned otherwise, all fine-tuning experiments run at 384 resolution (running fine-tuning at different resolution than training is common practice (Kolesnikov et al., 2020)).

When transferring ViT models to another dataset, we remove the whole head (two linear layers) and replace it by a single, zero-initialized linear layer outputting the number of classes required by the target dataset. We found this to be a little more robust than simply re-initializing the very last layer.

For VTAB we follow the protocol in Kolesnikov et al. (2020), and use the same hyperparameter setting for all tasks. We use a learning rate of 0.01 and train for 2500 steps (Tab. 4). We chose this setting by running a small sweep over two learning rates and two schedules, and selecting the setting with the highest VTAB score on the 200-example validation sets. We follow the pre-processing used in Kolesnikov et al. (2020), except that we do not use task-specific input resolutions. Instead we find that Vision Transformer benefits most from a high resolution (384×384) for all tasks.

B.1.2 SELF-SUPERVISION

We employ the *masked patch prediction* objective for preliminary self-supervision experiments. To do so we corrupt 50% of patch embeddings by either replacing their embeddings with a learnable [mask] embedding (80%), a random other patch embedding (10%) or just keeping them as is (10%). This setup is very similar to the one used for language by Devlin et al. (2019). Finally, we predict the 3-bit, mean color (i.e., 512 colors in total) of every corrupted patch using their respective patch representations.

We trained our self-supervised model for 1M steps (ca. 14 epochs) with batch size 4096 on JFT. We use Adam, with a base learning rate of $2 \cdot 10^{-4}$, warmup of 10k steps and cosine learning rate decay. As prediction targets for pretraining we tried the following settings: 1) predicting only the mean, 3bit color (i.e., 1 prediction of 512 colors), 2) predicting a 4×4 downsized version of the 16×16 patch with 3bit colors in parallel (i.e., 16 predictions of 512 colors), 3) regression on the full patch using L2 (i.e., 256 regressions on the 3 RGB channels). Surprisingly, we found that all worked quite well, though L2 was slightly worse. We report final results only for option 1) because it has shown best few-shot performance. We also experimented with 15% corruption rate as used by Devlin et al. (2019) but results were also slightly worse on our few-shot metrics.

Lastly, we would like to remark that our instantiation of masked patch prediction doesn’t require such an enormous amount of pretraining nor a large dataset such as JFT in order to lead to similar performance gains on ImageNet classification. That is, we observed diminishing returns on downstream performance after 100k pretraining steps, and see similar gains when pretraining on ImageNet.

C ADDITIONAL RESULTS

We report detailed results corresponding to the figures presented in the paper. Table 5 corresponds to Figure 3 from the paper and shows transfer performance of different ViT models pre-trained on datasets of increasing size: ImageNet, ImageNet-21k, and JFT-300M. Table 6 corresponds to

Dataset	Steps	Base LR
ImageNet	20 000	{0.003, 0.01, 0.03, 0.06}
CIFAR100	10 000	{0.001, 0.003, 0.01, 0.03}
CIFAR10	10 000	{0.001, 0.003, 0.01, 0.03}
Oxford-IIIT Pets	500	{0.001, 0.003, 0.01, 0.03}
Oxford Flowers-102	500	{0.001, 0.003, 0.01, 0.03}
VTAB (19 tasks)	2 500	0.01

表4: 微调超参数。所有模型均采用余弦学习率衰减进行微调, 批次大小为512, 无权重衰减, 梯度裁剪的全局范数为1。除非另有说明, 微调分辨率为384。

对于ResNets, 我们也运行了Kolesnikov等人(2020)的设置, 并在此次运行与我们的参数扫描中选取了最佳结果。最后, 除非另有说明, 所有微调实验均在384分辨率下进行(在不同于训练分辨率下进行微调是常见做法(Kolesnikov等人, 2020))。

将ViT模型迁移到另一个数据集时, 我们会移除整个头部(两个线性层), 并替换为一个单一、零初始化的线性层, 该层输出目标数据集所需的类别数量。我们发现这种方法比简单地重新初始化最后一层要稍微更稳健一些。

对于VTAB, 我们遵循Kolesnikov等人(2020)中的方案, 并对所有任务使用相同的超参数设置。我们采用0.01的学习率并训练2500步(表4)。该设置是通过两种学习率和两种训练计划进行小范围扫描, 并选择在200个样本的验证集上获得最高VTAB得分的配置而确定的。我们沿用Kolesnikov等人(2020)中的预处理方法, 但未采用任务特定的输入分辨率。相反, 我们发现视觉Transformer在所有任务中均能最高效地受益于高分辨率(384×384)输入。

B.1.2 自监督学习

我们采用*masked patch prediction*目标进行初步的自监督实验。为此, 我们通过以下方式破坏50%的补丁嵌入: 将其嵌入替换为可学习的[mask]嵌入(80%)、随机其他补丁嵌入(10%)或保持原样(10%)。这一设置与Devlin等人(2019)在语言任务中使用的方案非常相似。最后, 我们利用各补丁的表征预测每个被破坏补丁的3位平均颜色(即总计512种颜色)。

我们在JFT数据集上以4096的批次大小训练了我们的自监督模型100万步(约14个周期)。我们使用Adam优化器, 基础学习率为 2×10^{-4} , 预热步数为万步, 并采用余弦学习率衰减。在预训练的目标预测方面, 我们尝试了以下设置: 1) 仅预测均值、3位色彩(即对512种颜色进行1次预测); 2) 并行预测 16×16 图像块经 4×4 下采样后的3位色彩版本(即对512种颜色进行16次预测); 3) 使用L2损失对完整图像块进行回归(即对3个RGB通道进行256次回归)。令人惊讶的是, 我们发现所有方法都表现良好, 尽管L2损失稍逊一筹。我们仅报告选项1)的最终结果, 因为它在少样本学习任务中表现出最佳性能。我们还尝试了Devlin等人(2019)使用的15%掩蔽率, 但在我们的少样本评估指标上结果也略差。

最后, 我们想指出, 我们实现的掩码补丁预测不需要如此大量的预训练, 也不需要像JFT那样庞大的数据集, 就能在ImageNet分类任务上带来类似的性能提升。也就是说, 我们观察到在10万步预训练后, 下游性能的提升逐渐趋于平缓, 并且仅在ImageNet上进行预训练也能获得相似的收益。

C 附加结果

我们报告了与论文中图表相对应的详细结果。表5对应论文中的图3, 展示了在不同规模数据集(ImageNet、ImageNet-21k和JFT-300M)上预训练的各类ViT模型的迁移性能。表6则对应

		ViT-B/16	ViT-B/32	ViT-L/16	ViT-L/32	ViT-H/14
ImageNet	CIFAR-10	98.13	97.77	97.86	97.94	-
	CIFAR-100	87.13	86.31	86.35	87.07	-
	ImageNet	77.91	73.38	76.53	71.16	-
	ImageNet Real	83.57	79.56	82.19	77.83	-
	Oxford Flowers-102	89.49	85.43	89.66	86.36	-
	Oxford-IIIT-Pets	93.81	92.04	93.64	91.35	-
ImageNet-21k	CIFAR-10	98.95	98.79	99.16	99.13	99.27
	CIFAR-100	91.67	91.97	93.44	93.04	93.82
	ImageNet	83.97	81.28	85.15	80.99	85.13
	ImageNet Real	88.35	86.63	88.40	85.65	88.70
	Oxford Flowers-102	99.38	99.11	99.61	99.19	99.51
	Oxford-IIIT-Pets	94.43	93.02	94.73	93.09	94.82
JFT-300M	CIFAR-10	99.00	98.61	99.38	99.19	99.50
	CIFAR-100	91.87	90.49	94.04	92.52	94.55
	ImageNet	84.15	80.73	87.12	84.37	88.04
	ImageNet Real	88.85	86.27	89.99	88.28	90.33
	Oxford Flowers-102	99.56	99.27	99.56	99.45	99.68
	Oxford-IIIT-Pets	95.80	93.40	97.11	95.83	97.56

Table 5: Top1 accuracy (in %) of Vision Transformer on various datasets when pre-trained on ImageNet, ImageNet-21k or JFT300M. These values correspond to Figure 3 in the main text. Models are fine-tuned at 384 resolution. Note that the ImageNet results are computed without additional techniques (Polyak averaging and 512 resolution images) used to achieve results in Table 2.

name	Epochs	ImageNet	ImageNet Real	CIFAR-10	CIFAR-100	Pets	Flowers	exaFLOPs
ViT-B/32	7	80.73	86.27	98.61	90.49	93.40	99.27	55
ViT-B/16	7	84.15	88.85	99.00	91.87	95.80	99.56	224
ViT-L/32	7	84.37	88.28	99.19	92.52	95.83	99.45	196
ViT-L/16	7	86.30	89.43	99.38	93.46	96.81	99.66	783
ViT-L/16	14	87.12	89.99	99.38	94.04	97.11	99.56	1567
ViT-H/14	14	88.08	90.36	99.50	94.71	97.11	99.71	4262
ResNet50x1	7	77.54	84.56	97.67	86.07	91.11	94.26	50
ResNet50x2	7	82.12	87.94	98.29	89.20	93.43	97.02	199
ResNet101x1	7	80.67	87.07	98.48	89.17	94.08	95.95	96
ResNet152x1	7	81.88	87.96	98.82	90.22	94.17	96.94	141
ResNet152x2	7	84.97	89.69	99.06	92.05	95.37	98.62	563
ResNet152x2	14	85.56	89.89	99.24	91.92	95.75	98.75	1126
ResNet200x3	14	87.22	90.15	99.34	93.53	96.32	99.04	3306
R50x1+ViT-B/32	7	84.90	89.15	99.01	92.24	95.75	99.46	106
R50x1+ViT-B/16	7	85.58	89.65	99.14	92.63	96.65	99.40	274
R50x1+ViT-L/32	7	85.68	89.04	99.24	92.93	96.97	99.43	246
R50x1+ViT-L/16	7	86.60	89.72	99.18	93.64	97.03	99.40	859
R50x1+ViT-L/16	14	87.12	89.76	99.31	93.89	97.36	99.11	1668

Table 6: Detailed results of model scaling experiments. These correspond to Figure 5 in the main paper. We show transfer accuracy on several datasets, as well as the pre-training compute (in exaFLOPs).

Figure 5 from the paper and shows the transfer performance of ViT, ResNet, and hybrid models of varying size, as well as the estimated computational cost of their pre-training.

D ADDITIONAL ANALYSES

D.1 SGD VS. ADAM FOR RESNETS

ResNets are typically trained with SGD and our use of Adam as optimizer is quite unconventional. Here we show the experiments that motivated this choice. Namely, we compare the fine-tuning

		ViT-B/16	ViT-B/32	ViT-L/16	ViT-L/32	ViT-H/14
ImageNet	CIFAR-10	98.13	97.77	97.86	97.94	-
	CIFAR-100	87.13	86.31	86.35	87.07	-
	ImageNet	77.91	73.38	76.53	71.16	-
	ImageNet Real	83.57	79.56	82.19	77.83	-
	Oxford Flowers-102	89.49	85.43	89.66	86.36	-
	Oxford-IIIT-Pets	93.81	92.04	93.64	91.35	-
ImageNet-21k	CIFAR-10	98.95	98.79	99.16	99.13	99.27
	CIFAR-100	91.67	91.97	93.44	93.04	93.82
	ImageNet	83.97	81.28	85.15	80.99	85.13
	ImageNet Real	88.35	86.63	88.40	85.65	88.70
	Oxford Flowers-102	99.38	99.11	99.61	99.19	99.51
	Oxford-IIIT-Pets	94.43	93.02	94.73	93.09	94.82
JFT-300M	CIFAR-10	99.00	98.61	99.38	99.19	99.50
	CIFAR-100	91.87	90.49	94.04	92.52	94.55
	ImageNet	84.15	80.73	87.12	84.37	88.04
	ImageNet Real	88.85	86.27	89.99	88.28	90.33
	Oxford Flowers-102	99.56	99.27	99.56	99.45	99.68
	Oxford-IIIT-Pets	95.80	93.40	97.11	95.83	97.56

表5: 在ImageNet、ImageNet-21k或JFT300M上预训练后, Vision Transformer在不同数据集上的Top1准确率(%)。这些数值对应正文中的图3。模型均在384分辨率下进行微调。请注意, ImageNet结果的计算未采用表2中为达到最佳结果所使用的额外技术(Polyak平均和512分辨率图像)。

name	Epochs	ImageNet	ImageNet Real	CIFAR-10	CIFAR-100	Pets	Flowers	exaFLOPs
ViT-B/32	7	80.73	86.27	98.61	90.49	93.40	99.27	55
ViT-B/16	7	84.15	88.85	99.00	91.87	95.80	99.56	224
ViT-L/32	7	84.37	88.28	99.19	92.52	95.83	99.45	196
ViT-L/16	7	86.30	89.43	99.38	93.46	96.81	99.66	783
ViT-L/16	14	87.12	89.99	99.38	94.04	97.11	99.56	1567
ViT-H/14	14	88.08	90.36	99.50	94.71	97.11	99.71	4262
ResNet50x1	7	77.54	84.56	97.67	86.07	91.11	94.26	50
ResNet50x2	7	82.12	87.94	98.29	89.20	93.43	97.02	199
ResNet101x1	7	80.67	87.07	98.48	89.17	94.08	95.95	96
ResNet152x1	7	81.88	87.96	98.82	90.22	94.17	96.94	141
ResNet152x2	7	84.97	89.69	99.06	92.05	95.37	98.62	563
ResNet152x2	14	85.56	89.89	99.24	91.92	95.75	98.75	1126
ResNet200x3	14	87.22	90.15	99.34	93.53	96.32	99.04	3306
R50x1+ViT-B/32	7	84.90	89.15	99.01	92.24	95.75	99.46	106
R50x1+ViT-B/16	7	85.58	89.65	99.14	92.63	96.65	99.40	274
R50x1+ViT-L/32	7	85.68	89.04	99.24	92.93	96.97	99.43	246
R50x1+ViT-L/16	7	86.60	89.72	99.18	93.64	97.03	99.40	859
R50x1+ViT-L/16	14	87.12	89.76	99.31	93.89	97.36	99.11	1668

表6: 模型缩放实验的详细结果。这些对应主论文中的图5。我们展示了在多个数据集上的迁移准确率, 以及预训练计算量(单位: ex-aFLOPs)。

论文中的图5展示了不同规模的ViT、ResNet以及混合模型的迁移性能, 以及它们预训练的估计计算成本。

D 额外分析

D.1 残差网络中SGD与ADAM的对比

ResNets通常使用SGD进行训练, 而我们采用Adam作为优化器是相当非传统的。这里我们展示了促使这一选择的实验。具体而言, 我们比较了微调

Dataset	ResNet50		ResNet152x2	
	Adam	SGD	Adam	SGD
ImageNet	77.54	78.24	84.97	84.37
CIFAR10	97.67	97.46	99.06	99.07
CIFAR100	86.07	85.17	92.05	91.06
Oxford-IIIT Pets	91.11	91.00	95.37	94.79
Oxford Flowers-102	94.26	92.06	98.62	99.32
Average	89.33	88.79	94.01	93.72

Table 7: Fine-tuning ResNet models pre-trained with Adam and SGD.

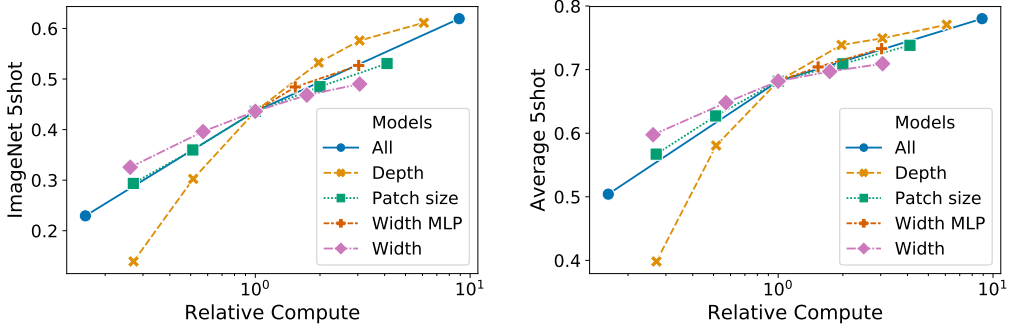


Figure 8: Scaling different model dimensions of the Vision Transformer.

performance of two ResNets – 50x1 and 152x2 – pre-trained on JFT with SGD and Adam. For SGD, we use the hyperparameters recommended by Kolesnikov et al. (2020). Results are presented in Table 7. Adam pre-training outperforms SGD pre-training on most datasets and on average. This justifies the choice of Adam as the optimizer used to pre-train ResNets on JFT. Note that the absolute numbers are lower than those reported by Kolesnikov et al. (2020), since we pre-train only for 7 epochs, not 30.

D.2 TRANSFORMER SHAPE

We ran ablations on scaling different dimensions of the Transformer architecture to find out which are best suited for scaling to very large models. Figure 8 shows 5-shot performance on ImageNet for different configurations. All configurations are based on a ViT model with 8 layers, $D = 1024$, $D_{MLP} = 2048$ and a patch size of 32, the intersection of all lines. We can see that scaling the depth results in the biggest improvements which are clearly visible up until 64 layers. However, diminishing returns are already visible after 16 layers. Interestingly, scaling the width of the network seems to result in the smallest changes. Decreasing the patch size and thus increasing the effective sequence length shows surprisingly robust improvements without introducing parameters. These findings suggest that compute might be a better predictor of performance than the number of parameters, and that scaling should emphasize depth over width if any. Overall, we find that scaling all dimensions proportionally results in robust improvements.

D.3 HEAD TYPE AND CLASS TOKEN

In order to stay as close as possible to the original Transformer model, we made use of an additional [class] token, which is taken as image representation. The output of this token is then transformed into a class prediction via a small multi-layer perceptron (MLP) with tanh as non-linearity in the single hidden layer.

This design is inherited from the Transformer model for text, and we use it throughout the main paper. An initial attempt at using only image-patch embeddings, globally average-pooling (GAP) them, followed by a linear classifier—just like ResNet’s final feature map—performed very poorly. However, we found that this is neither due to the extra token, nor to the GAP operation. Instead,

Dataset	ResNet50		ResNet152x2	
	Adam	SGD	Adam	SGD
ImageNet	77.54	78.24	84.97	84.37
CIFAR10	97.67	97.46	99.06	99.07
CIFAR100	86.07	85.17	92.05	91.06
Oxford-IIIT Pets	91.11	91.00	95.37	94.79
Oxford Flowers-102	94.26	92.06	98.62	99.32
Average	89.33	88.79	94.01	93.72

表7: 使用Adam和SGD预训练的ResNet模型微调结果。

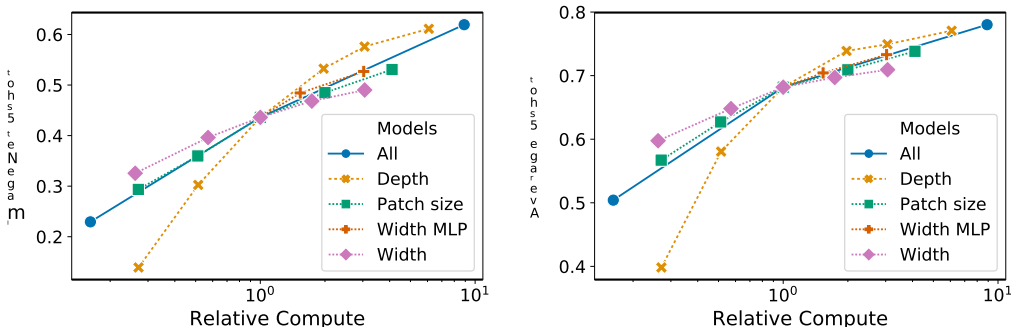


图8: 缩放视觉Transformer的不同模型维度。

在JFT上使用SGD和Adam预训练的两个ResNet——50x1和152x2——的性能表现。对于SGD，我们采用了Kolesnikov等人（2020年）推荐的超参数。结果如表7所示。在大多数数据集及平均表现上，Adam预训练均优于SGD预训练。这证明了选择Adam作为在JFT上预训练ResNet优化器的合理性。请注意，由于我们仅进行了7个周期（而非30个周期）的预训练，绝对数值低于Kolesnikov等人（2020年）报告的结果。

D.2 变压器形状

我们对Transformer架构的不同维度进行了缩放消融实验，以找出哪些维度最适合扩展到超大型模型。图8展示了不同配置在ImageNet上的5样本性能。所有配置均基于一个ViT模型，该模型具有8层、 $D = 1024$ 、 $D_{MLP} = 2048$ 以及32的补丁大小，即所有线条的交汇点。我们可以看到，增加深度带来的改进最大，直到64层都清晰可见。然而，在16层之后已经可见收益递减。有趣的是，增加网络宽度似乎带来的变化最小。减小补丁大小从而增加有效序列长度，出人意料地带来了稳健的改进，且没有引入额外参数。这些发现表明，计算量可能是比参数数量更好的性能预测指标，并且如果必须选择，缩放应优先考虑深度而非宽度。总体而言，我们发现按比例缩放所有维度能带来稳健的改进。

D.3 头部类型与类别标记

为了尽可能贴近原始Transformer模型，我们使用了一个额外的[class]标记作为图像表示。该标记的输出随后通过一个小型多层感知机（MLP）转换为类别预测，该感知机在单一隐藏层中使用tanh作为非线性激活函数。

这一设计继承自文本Transformer模型，我们在整篇主论文中都沿用了它。最初尝试仅使用图像块嵌入，对其进行全局平均池化（GAP），再接入线性分类器——类似ResNet最终特征图的做法——效果极差。但我们发现，这既非由额外标记引起，也非GAP操作所致。实际上，

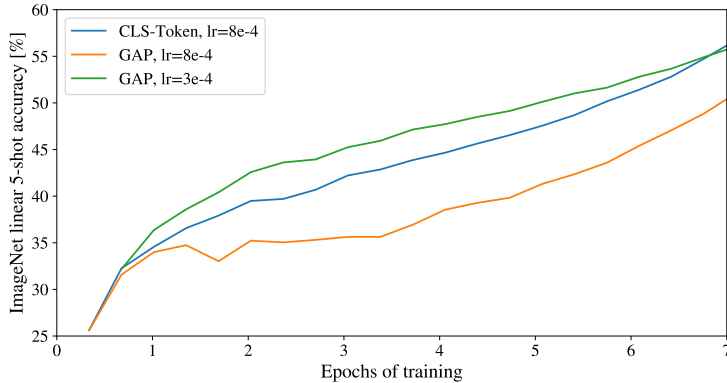


Figure 9: Comparison of class-token and global average pooling classifiers. Both work similarly well, but require different learning-rates.

Pos. Emb.	Default/Stem	Every Layer	Every Layer-Shared
No Pos. Emb.	0.61382	N/A	N/A
1-D Pos. Emb.	0.64206	0.63964	0.64292
2-D Pos. Emb.	0.64001	0.64046	0.64022
Rel. Pos. Emb.	0.64032	N/A	N/A

Table 8: Results of the ablation study on positional embeddings with ViT-B/16 model evaluated on ImageNet 5-shot linear.

the difference in performance is fully explained by the requirement for a different learning-rate, see Figure 9.

D.4 POSITIONAL EMBEDDING

We ran ablations on different ways of encoding spatial information using positional embedding. We tried the following cases:

- Providing no positional information: Considering the inputs as a *bag of patches*.
- 1-dimensional positional embedding: Considering the inputs as a sequence of patches in the raster order (default across all other experiments in this paper).
- 2-dimensional positional embedding: Considering the inputs as a grid of patches in two dimensions. In this case, two sets of embeddings are learned, each for one of the axes, X -embedding, and Y -embedding, each with size $D/2$. Then, based on the coordinate on the path in the input, we concatenate the X and Y embedding to get the final positional embedding for that patch.
- Relative positional embeddings: Considering the relative distance between patches to encode the spatial information as instead of their absolute position. To do so, we use 1-dimensional Relative Attention, in which we define the relative distance all possible pairs of patches. Thus, for every given pair (one as query, and the other as key/value in the attention mechanism), we have an offset $p_q - p_k$, where each offset is associated with an embedding. Then, we simply run extra attention, where we use the original query (the content of query), but use relative positional embeddings as keys. We then use the logits from the relative attention as a bias term and add it to the logits of the main attention (content-based attention) before applying the softmax.

In addition to different ways of encoding spatial information, we also tried different ways of incorporating this information in our model. For the 1-dimensional and 2-dimensional positional embeddings, we tried three different cases: (1) add positional embeddings to the inputs right after

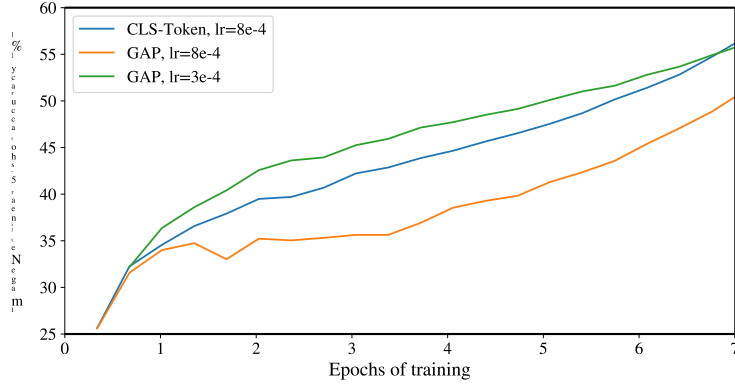


图9：类令牌与全局平均池化分类器的比较。两者表现相似，但需要不同的学习率。

Pos. Emb.	Default/Stem	Every Layer	Every Layer-Shared
No Pos. Emb.	0.61382	N/A	N/A
1-D Pos. Emb.	0.64206	0.63964	0.64292
2-D Pos. Emb.	0.64001	0.64046	0.64022
Rel. Pos. Emb.	0.64032	N/A	N/A

表8：使用ViT-B/16模型在ImageNet 5-shot线性评估中关于位置嵌入的消融研究结果。

性能差异完全由不同学习率的需求所解释，参见图9。

D.4 位置嵌入

我们通过位置嵌入对编码空间信息的不同方式进行了消融实验。我们尝试了以下情况：

- 不提供位置信息：将输入视为一个 *bag of patches*。
- 一维位置嵌入：将输入视为光栅顺序中的一系列补丁（本文所有其他实验的默认设置）。
- 二维位置嵌入：将输入视为二维的补丁网格。在这种情况下，学习两组嵌入，每组对应一个轴，即X嵌入和Y嵌入，每组大小为 $D/2$ 。然后，根据输入中路径的坐标，我们将X和Y嵌入连接起来，得到该补丁的最终位置嵌入。
- 相对位置嵌入：考虑图像块之间的相对距离来编码空间信息，而非使用绝对位置。为此，我们采用一维相对注意力机制，其中我们定义了所有可能图像块对之间的相对距离。因此，对于每一对给定的图像块（一个作为查询，另一个作为注意力机制中的键/值），我们有一个偏移量 $p_q - p_k$ ，每个偏移量都与一个嵌入向量相关联。接着，我们额外运行一次注意力计算，使用原始查询（查询的内容），但以相对位置嵌入作为键。然后，我们将相对注意力得到的逻辑值作为偏置项，在应用softmax之前将其添加到主注意力（基于内容的注意力）的逻辑值中。

除了编码空间信息的不同方式外，我们还尝试了在模型中融入这些信息的不同方法。对于一维和二维位置嵌入，我们尝试了三种不同情况：（1）在输入后直接添加位置嵌入

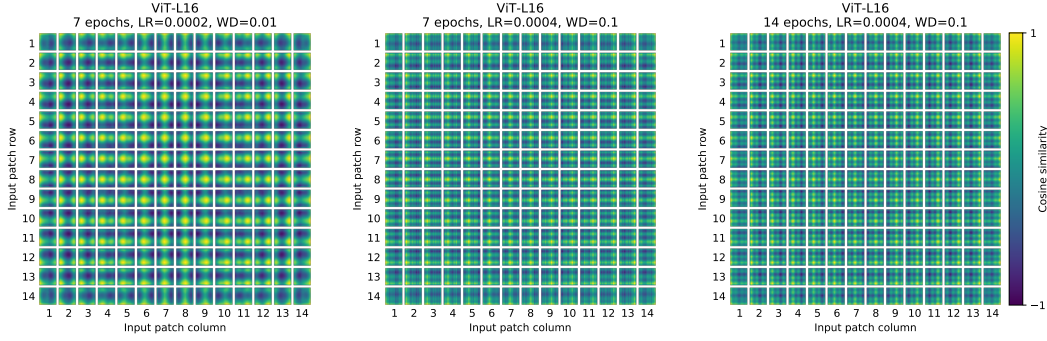


Figure 10: Position embeddings of models trained with different hyperparameters.

the stem of them model and before feeding the inputs to the Transformer encoder (default across all other experiments in this paper); (2) learn and add positional embeddings to the inputs at the beginning of each layer; (3) add a learned positional embeddings to the inputs at the beginning of each layer (shared between layers).

Table 8 summarizes the results from this ablation study on a ViT-B/16 model. As we can see, while there is a large gap between the performances of the model with no positional embedding and models with positional embedding, there is little to no difference between different ways of encoding positional information. We speculate that since our Transformer encoder operates on patch-level inputs, as opposed to pixel-level, the differences in how to encode spatial information is less important. More precisely, in patch-level inputs, the spatial dimensions are much smaller than the original pixel-level inputs, e.g., 14×14 as opposed to 224×224 , and learning to represent the spatial relations in this resolution is equally easy for these different positional encoding strategies. Even so, the specific pattern of position embedding similarity learned by the network depends on the training hyperparameters (Figure 10).

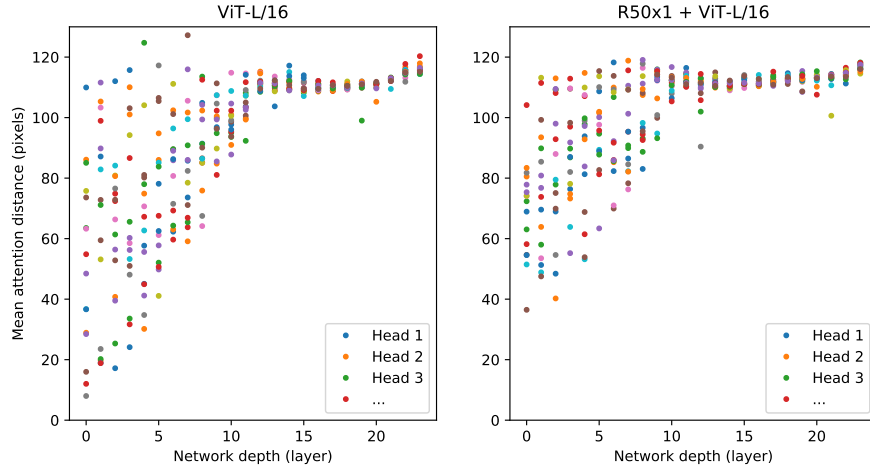


Figure 11: Size of attended area by head and network depth. Attention distance was computed for 128 example images by averaging the distance between the query pixel and all other pixels, weighted by the attention weight. Each dot shows the mean attention distance across images for one of 16 heads at one layer. Image width is 224 pixels.

D.5 EMPIRICAL COMPUTATIONAL COSTS

We are also interested in real-world speed of the architectures on our hardware, which is not always well predicted by theoretical FLOPs due to details like lane widths and cache sizes. For this purpose,

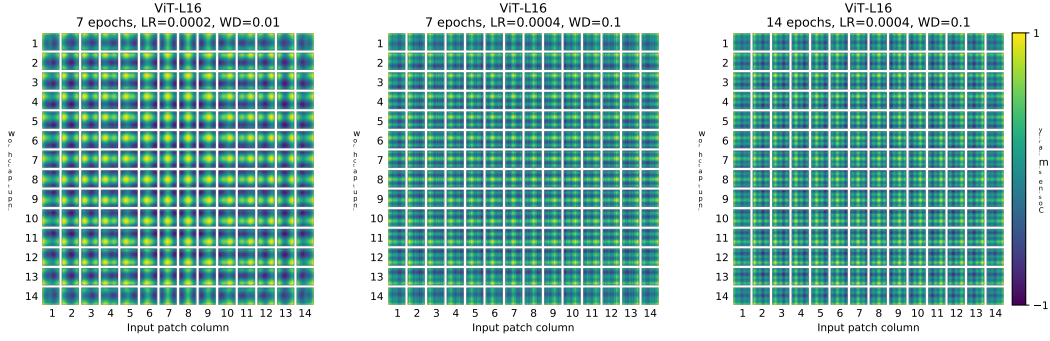


图 10: 使用不同超参数训练的模型的位置嵌入参数。

在将它们模型的词干输入Transformer编码器之前（本文所有其他实验的默认设置）；(2) 在每一层开始时学习并向输入添加位置嵌入；(3) 在每一层开始时向输入添加一个学习到的位置嵌入（各层之间共享）。

表8总结了在ViT-B/16模型上进行的消融研究结果。可以看出，虽然无位置嵌入的模型与带有位置嵌入的模型之间存在较大性能差距，但不同位置信息编码方式之间几乎没有差异。我们推测，由于我们的Transformer编码器处理的是图像块级别的输入（而非像素级别），如何编码空间信息的差异变得不那么重要。更准确地说，在图像块级别的输入中，空间维度远小于原始像素级别的输入（例如 14×14 对比 224×224 ），在此分辨率下学习表示空间关系对于这些不同的位置编码策略同样容易。即便如此，网络学习到的位置嵌入相似性的具体模式仍取决于训练超参数（图10）。

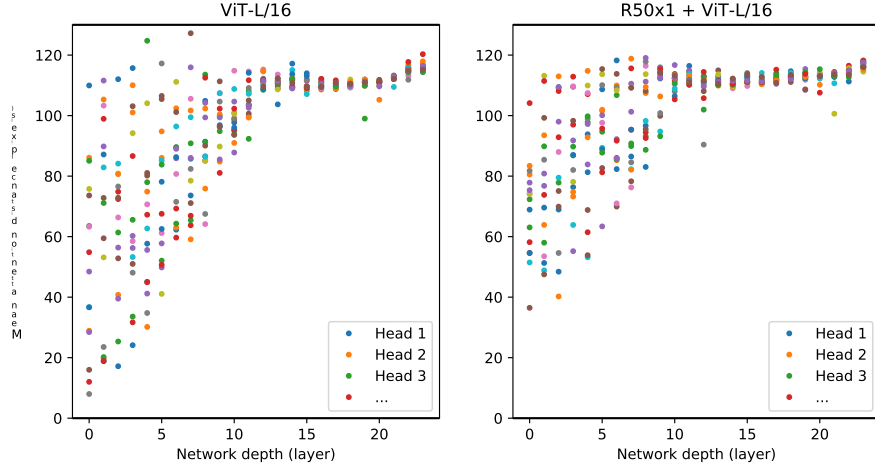


图11: 按头和网络深度划分的关注区域大小。通过计算查询像素与所有其他像素之间的距离（以注意力权重加权平均），对128张示例图像进行了注意力距离的计算。每个点表示某一层中16个头之一在所有图像上的平均注意力距离。图像宽度为224像素。

D.5 实证计算成本

我们还对架构在我们硬件上的实际速度感兴趣，由于通道宽度和缓存大小等细节，理论浮点运算次数并不总能准确预测实际速度。为此，

we perform timing of inference speed for the main models of interest, on a TPUv3 accelerator; the difference between inference and backprop speed is a constant model-independent factor.

Figure 12 (left) shows how many images one core can handle per second, across various input sizes. Every single point refers to the peak performance measured across a wide range of batch-sizes. As can be seen, the theoretical bi-quadratic scaling of ViT with image size only barely starts happening for the largest models at the largest resolutions.

Another quantity of interest is the largest batch-size each model can fit onto a core, larger being better for scaling to large datasets. Figure 12 (right) shows this quantity for the same set of models. This shows that large ViT models have a clear advantage in terms of memory-efficiency over ResNet models.

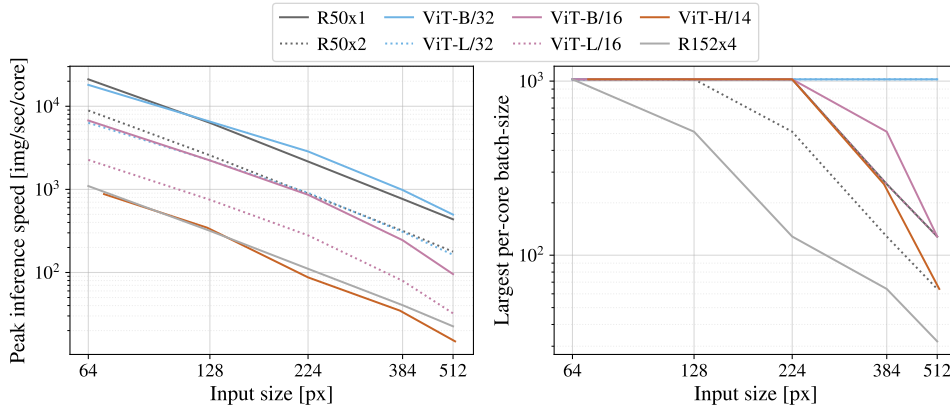


Figure 12: **Left:** Real wall-clock timings of various architectures across input sizes. ViT models have speed comparable to similar ResNets. **Right:** Largest per-core batch-size fitting on device with various architectures across input sizes. ViT models are clearly more memory-efficient.

D.6 AXIAL ATTENTION

Axial Attention (Huang et al., 2020; Ho et al., 2019) is a simple, yet effective technique to run self-attention on large inputs that are organized as multidimensional tensors. The general idea of axial attention is to perform multiple attention operations, each along a single axis of the input tensor, instead of applying 1-dimensional attention to the flattened version of the input. In axial attention, each attention mixes information along a particular axis, while keeping information along the other axes independent. Along this line, Wang et al. (2020b) proposed the AxialResNet model in which all the convolutions with kernel size 3×3 in a ResNet50 are replaced by axial self-attention, i.e. a row and column attention, augmented by relative positional encoding. We have implemented AxialResNet as a baseline model.³

Moreover, we have modified ViT to process inputs in the 2-dimensional shape, instead of a 1-dimensional sequence of patches, and incorporate Axial Transformer blocks, in which instead of a self-attention followed by an MLP, we have a row-self-attention plus an MLP followed by a column-self-attention plus an MLP.

Figure 13, present the performance of Axial ResNet, Axial-ViT-B/32 and Axial-ViT-B/16 on ImageNet 5shot linear, when pretrained on JFT dataset, verses the pretraining compute, both in terms of number of FLOPs and inference time (example per seconds). As we can see, both Axial-ViT-B/32 and Axial-ViT-B/16 do better than their ViT-B counterpart in terms of performance, but it comes at

³Our implementation is based on the open-sourced PyTorch implementation in <https://github.com/cshrddlam/axial-deeplab>. In our experiments, we reproduced the scores reported in (Wang et al., 2020b) in terms of accuracy, however, our implementation, similar to the open-source implementation, is very slow on TPUs. Therefore, we were not able to use it for extensive large-scale experiments. These may be unlocked by a carefully optimized implementation.

我们在TPUv3加速器上对主要关注模型进行了推理速度的计时；推理与反向传播速度之间的差异是一个与模型无关的恒定因子。

图12（左）展示了单个核心每秒能处理多少张图像，覆盖了各种输入尺寸。每个点都代表了在广泛批量大小范围内测得的峰值性能。可以看出，ViT理论上的双二次缩放与图像大小的关系，仅在最大分辨率下的最大模型中才刚刚开始显现。

另一个值得关注的量是每个模型能在一个核心上容纳的最大批次大小，这个值越大，越有利于扩展到大型数据集。图12（右）展示了同一组模型的这一数值。这表明，在内存效率方面，大型ViT模型相比ResNet模型具有明显优势。

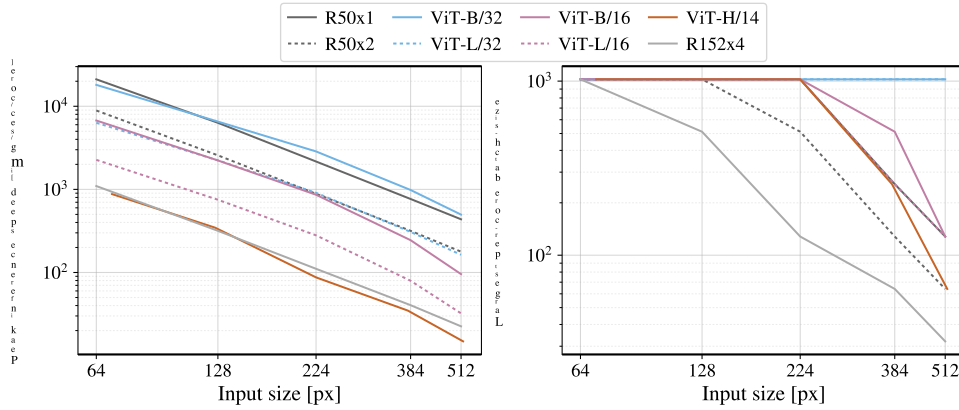


图12：左：不同架构在不同输入尺寸下的实际挂钟时间。ViT模型的速度与类似的ResNet相当。右：不同架构在不同输入尺寸下设备上能容纳的最大单核批次大小。ViT模型明显更具内存效率。

D.6 轴向注意力

轴向注意力（Huang等人，2020；Ho等人，2019）是一种简单而有效的技术，用于对以多维张量形式组织的大型输入进行自注意力计算。轴向注意力的核心思想是执行多次注意力操作，每次沿输入张量的单个轴进行，而不是将输入展平后应用一维注意力。在轴向注意力中，每次注意力操作会沿特定轴混合信息，同时保持其他轴的信息相互独立。沿着这一思路，Wang等人（2020b）提出了AxialResNet模型，该模型将ResNet50中所有卷积核大小为 3×3 的卷积层替换为轴向自注意力（即行注意力和列注意力），并辅以相对位置编码。我们已将AxialResNet实现为基线模型。³

此外，我们对ViT进行了改进，使其能够处理二维形状的输入，而非一维的补丁序列，并引入了轴向Transformer模块。在该模块中，我们不再采用自注意力机制后接MLP的结构，而是采用行自注意力加MLP，再后接列自注意力加MLP的结构。

图13展示了Axial ResNet、Axial-ViT-B/32和Axial-ViT-B/16在JFT数据集上预训练后，于ImageNet 5-shot线性评估中的性能表现，对比了预训练计算量（以FLOPs和每秒推理样本数为衡量指标）。如图所示，Axial-ViT-B/32和Axial-ViT-B/16在性能上均优于对应的ViT-B模型，但这一优势伴随着

³Our implementation is based on the open-sourced PyTorch implementation in <https://github.com/csrhddlam/axial-deeplab>. In our experiments, we reproduced the scores reported in (Wang et al., 2020b) in terms of accuracy, however, our implementation, similar to the open-source implementation, is very slow on TPUs. Therefore, we were not able to use it for extensive large-scale experiments. These may be unlocked by a carefully optimized implementation.

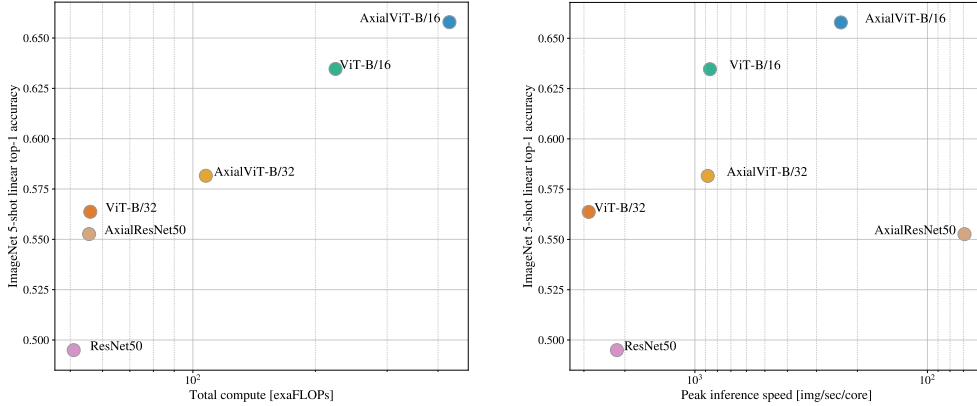


Figure 13: Performance of Axial-Attention based models, in terms of top-1 accuracy on ImageNet 5-shot linear, versus their speed in terms of number of FLOPs (**left**) and inference time (**left**).

the cost of more compute. This is because in Axial-ViT models, each Transformer block with global self-attention is replaced by two Axial Transformer blocks, one with row and one with column self-attention and although the sequence length that self-attention operates on is smaller in axial case, there is a extra MLP per Axial-ViT block. For the AxialResNet, although it looks reasonable in terms of accuracy/compute trade-off (Figure 13, left), the naive implementation is extremely slow on TPUs (Figure 13, right).

D.7 ATTENTION DISTANCE

To understand how ViT uses self-attention to integrate information across the image, we analyzed the average distance spanned by attention weights at different layers (Figure 11). This “attention distance” is analogous to receptive field size in CNNs. Average attention distance is highly variable across heads in lower layers, with some heads attending to much of the image, while others attend to small regions at or near the query location. As depth increases, attention distance increases for all heads. In the second half of the network, most heads attend widely across tokens.

D.8 ATTENTION MAPS

To compute maps of the attention from the output token to the input space (Figures 6 and 14), we used Attention Rollout (Abnar & Zuidema, 2020). Briefly, we averaged attention weights of ViT-L/16 across all heads and then recursively multiplied the weight matrices of all layers. This accounts for the mixing of attention across tokens through all layers.

D.9 OBJECTNET RESULTS

We also evaluate our flagship ViT-H/14 model on the ObjectNet benchmark following the evaluation setup in Kolesnikov et al. (2020), resulting in 82.1% top-5 accuracy and 61.7% top-1 accuracy.

D.10 VTAB BREAKDOWN

Table 9 shows the scores attained on each of the VTAB-1k tasks.

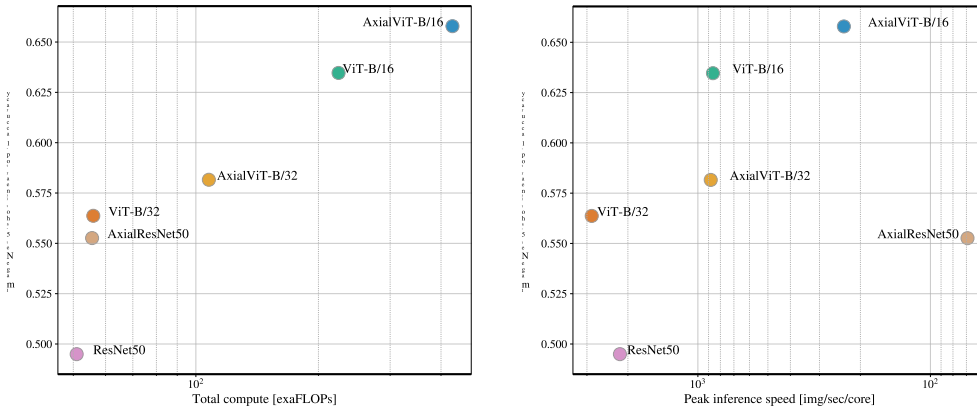


图13: 基于轴向注意力模型的性能, 以ImageNet 5-shot线性分类的top-1准确率为衡量标准, 与其在FLOPs数量 (左) 和推理时间 (左) 方面的速度对比。

更高的计算成本。这是因为在轴向视觉Transformer模型中, 每个采用全局自注意力的Transformer模块被两个轴向Transformer模块取代——一个采用行自注意力, 另一个采用列自注意力。尽管在轴向模型中自注意力操作的序列长度更短, 但每个轴向视觉Transformer模块都额外增加了一个多层感知机。对于轴向残差网络而言, 虽然其在准确率与计算成本的权衡方面看似合理 (图13左), 但其朴素实现在TPU上的运行速度极慢 (图13右)。

D.7 注意力距离

为了理解ViT如何利用自注意力机制整合图像信息, 我们分析了不同层中注意力权重跨越的平均距离 (图11)。这种“注意力距离”类似于CNN中的感受野大小。在较低层中, 不同注意力头的平均注意力距离差异很大: 有些注意力头关注图像的大部分区域, 而另一些则仅关注查询位置附近的小范围区域。随着网络深度增加, 所有注意力头的注意力距离均逐渐增大。在网络的后半部分, 大多数注意力头会在所有标记之间进行广泛关注。

D.8 注意力图谱

为了计算从输出标记到输入空间的注意力图 (图6和图14), 我们采用了注意力展开方法 (Abnar & Zuidema, 2020)。简而言之, 我们平均了ViT-L/16所有注意力头的权重, 然后递归地乘以所有层的权重矩阵。这考虑了所有层中标记间注意力的混合效应。

D.9 OBJECTNET 结果

我们还在ObjectNet基准上评估了我们的旗舰模型ViT-H/14, 遵循Kolesnikov等人 (2020) 的评估设置, 取得了82.1%的top-5准确率和61.7%的top-1准确率。

D.10 VTAB 细分

表9显示了在VTAB-1k各项任务上取得的分数。

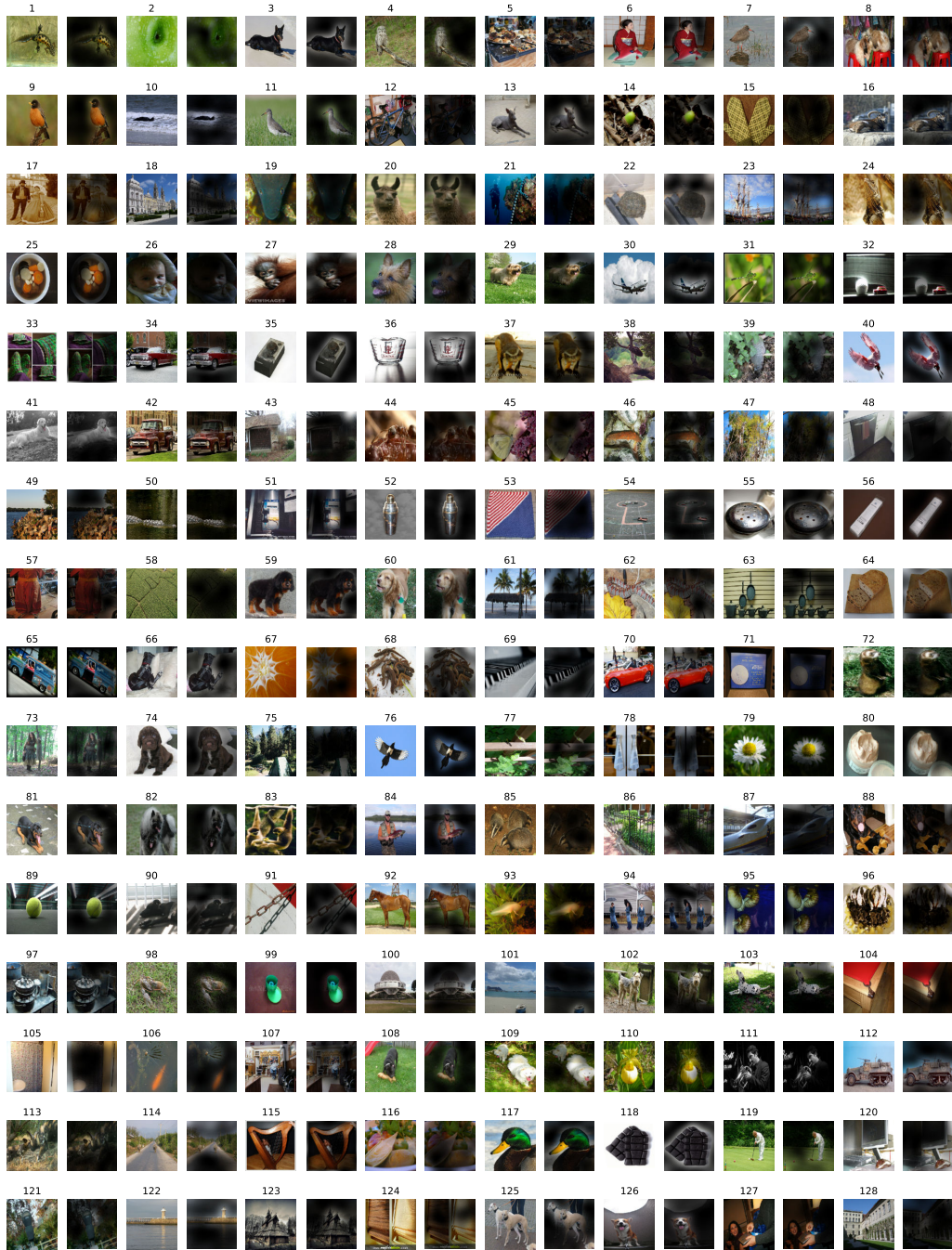


Figure 14: Further example attention maps as in Figure 6 (random selection).

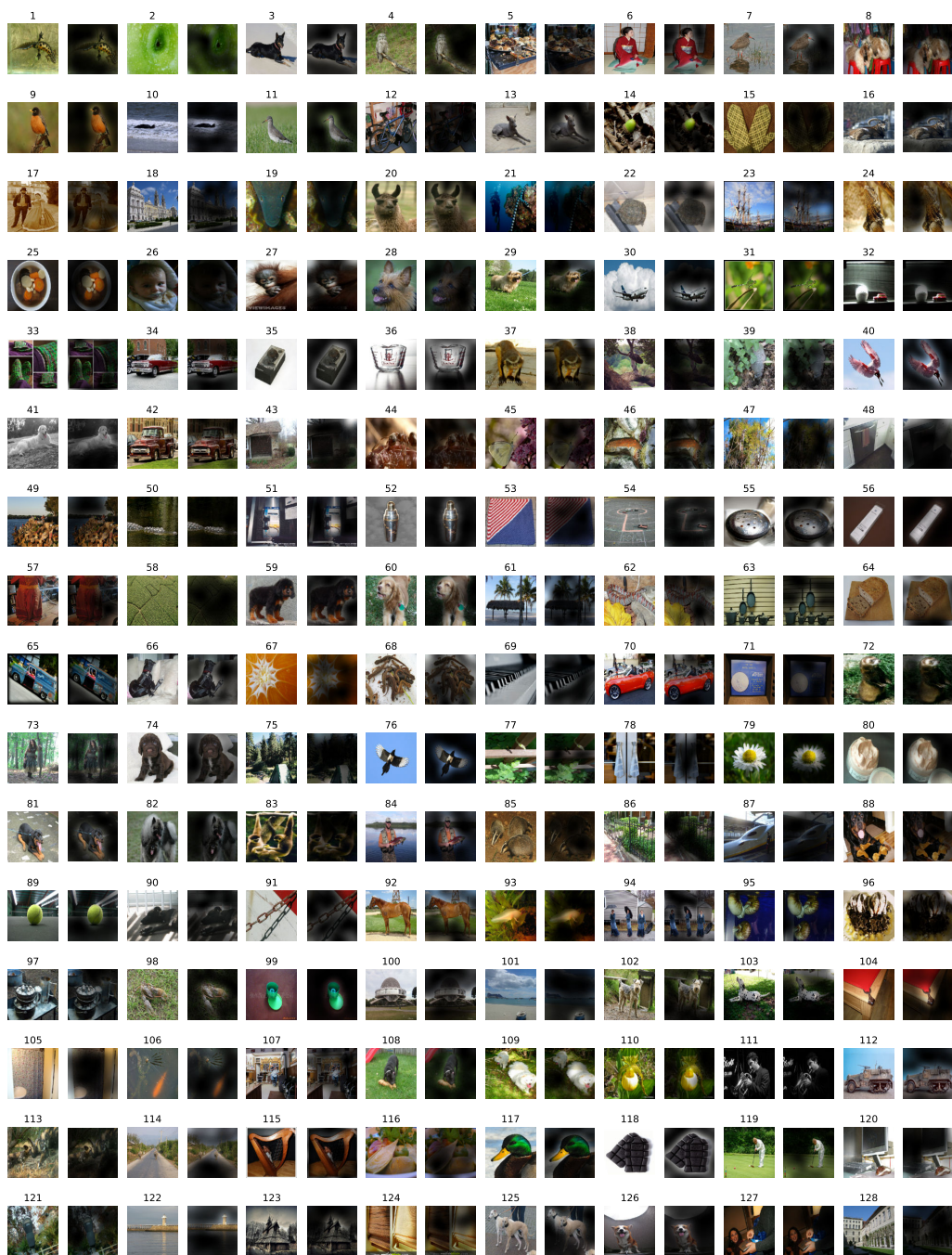


图14: 如图6所示的进一步示例注意力图 (随机选取)。

Table 9: Breakdown of VTAB-1k performance across tasks.

	● Caltech101	● CIFAR-100	● DTD	● Flowers102	● Pets	● Sun397	● SVHN	● Camelyon	● EuroSAT	● Resisc45	● Retinopathy	● Clevr-Count	● Clevr-Dist	● DMLab	● dSpr-Loc	● dSpr-Ori	● KITTI-Dist	● sNORB-Azim	● sNORB-Elev	● Mean
ViT-H/14 (JFT)	95.3	85.5	75.2	99.7	97.2	65.0	88.9	83.3	96.7	91.4	76.6	91.7	63.8	53.1	79.4	63.3	84.5	33.2	51.2	77.6
ViT-L/16 (JFT)	95.4	81.9	74.3	99.7	96.7	63.5	87.4	83.6	96.5	89.7	77.1	86.4	63.1	49.7	74.5	60.5	82.2	36.2	51.1	76.3
ViT-L/16 (I21k)	90.8	84.1	74.1	99.3	92.7	61.0	80.9	82.5	95.6	85.2	75.3	70.3	56.1	41.9	74.7	64.9	79.9	30.5	41.7	72.7

表9: VTAB-1k 在各任务中的性能细分。

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